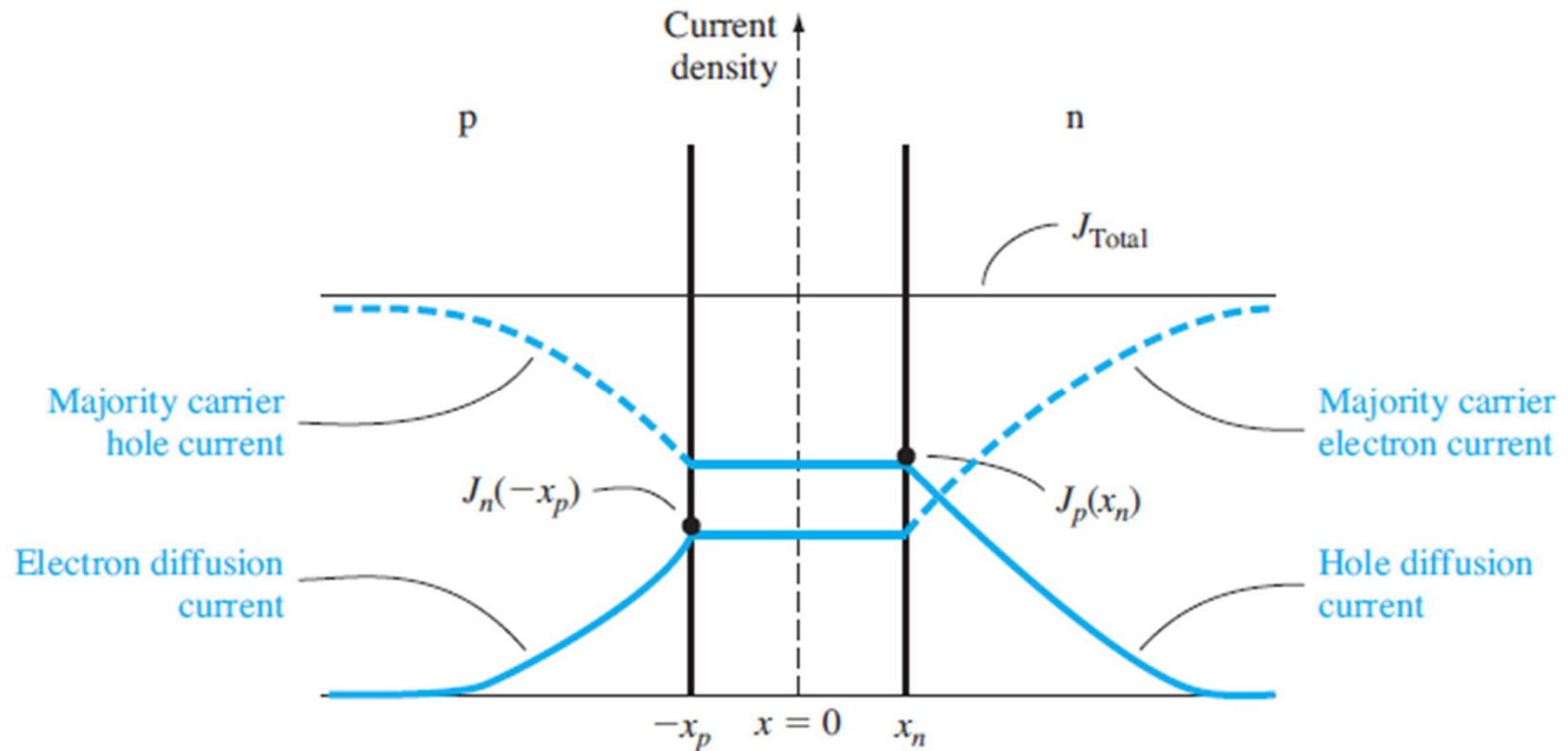


L8: The pn Junction Diode



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pn Junction under different bias

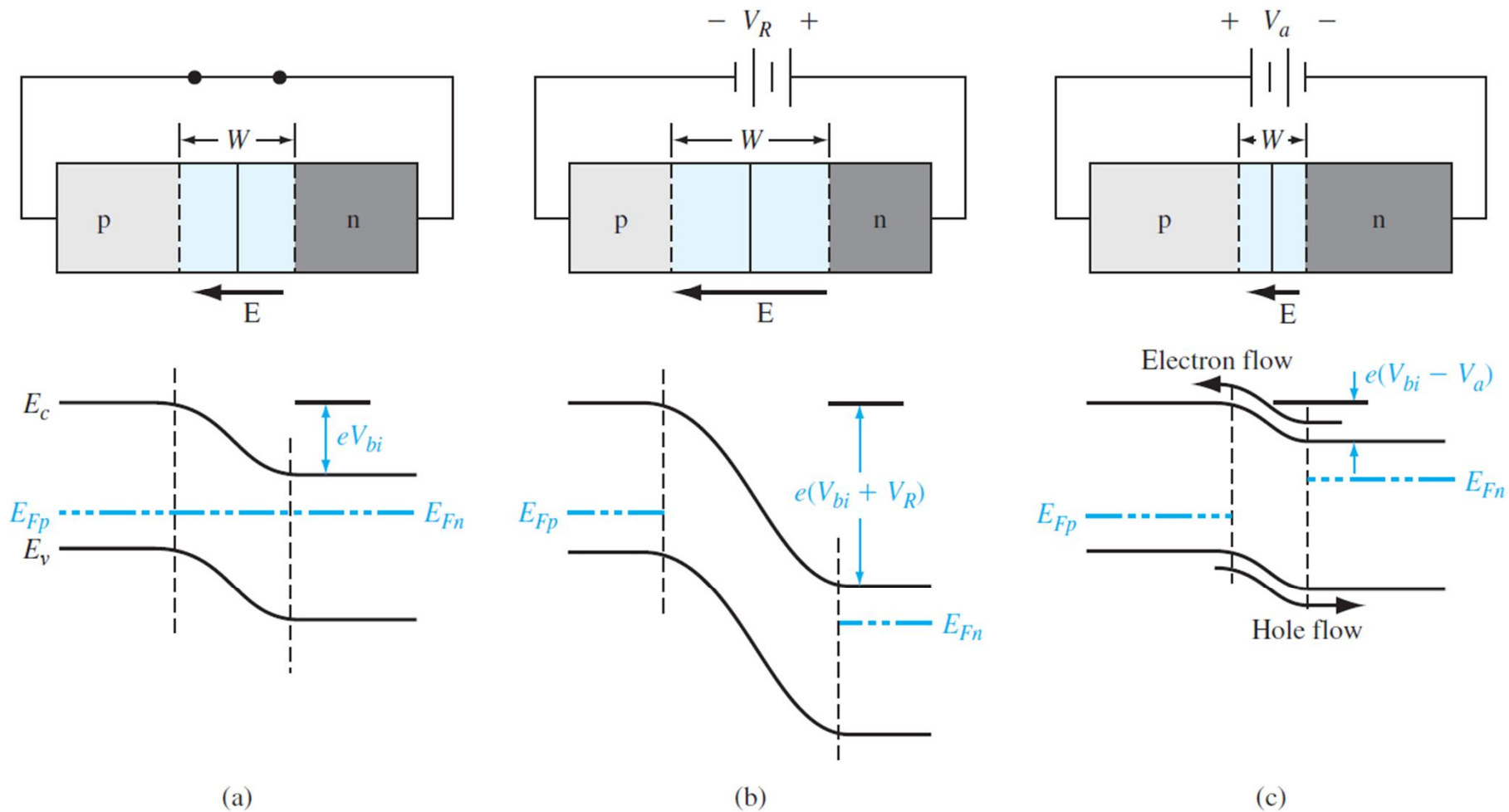
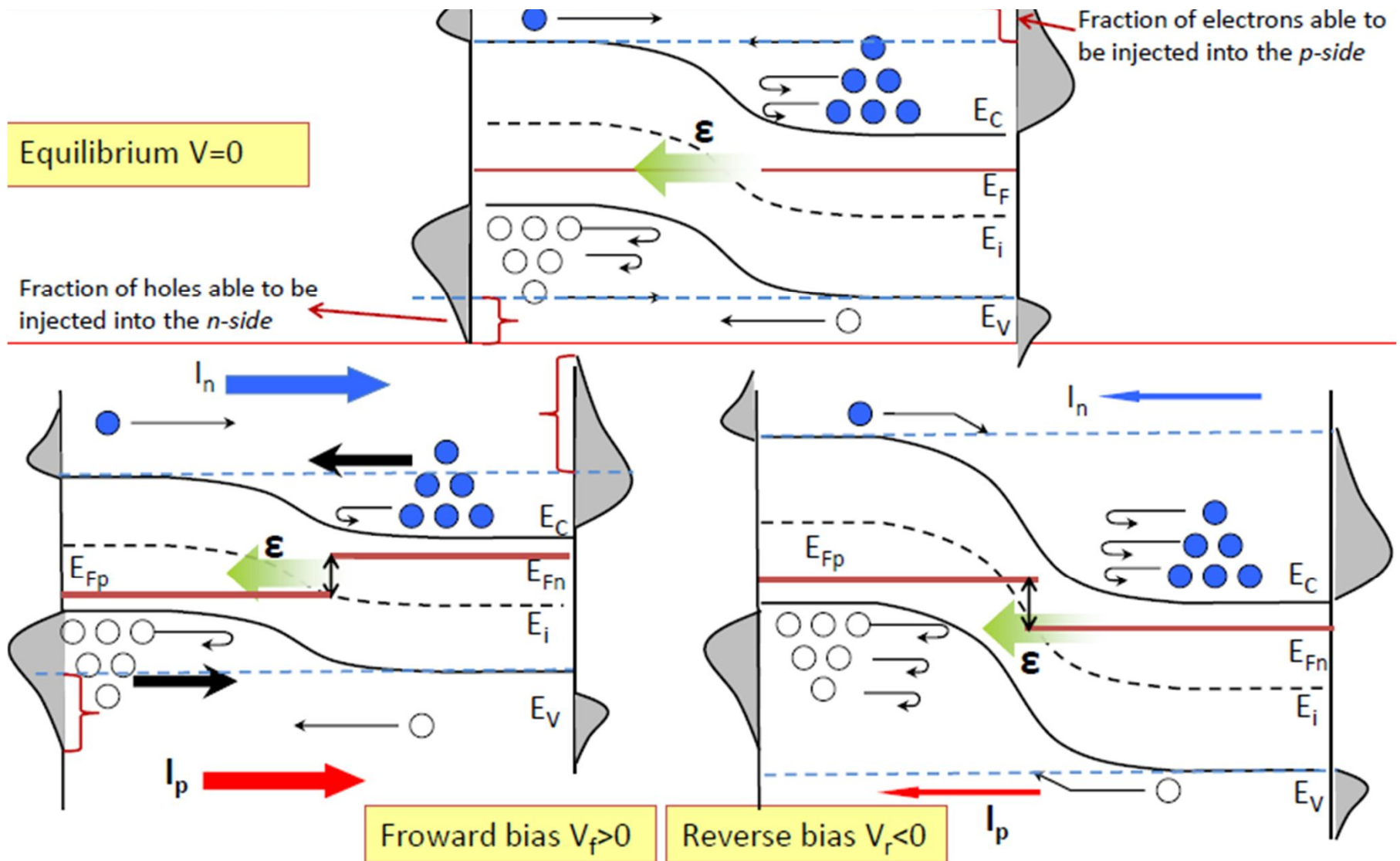


Figure 8.1 | A pn junction and its associated energy-band diagram for (a) zero bias, (b) reverse bias, and (c) forward bias.

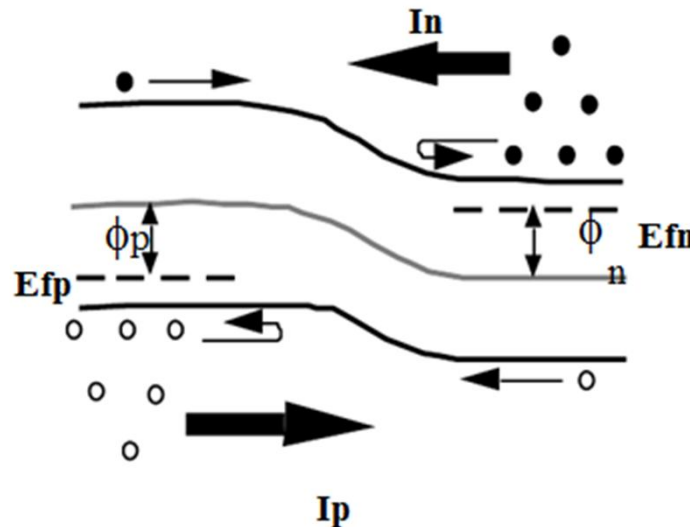
pn Junction under different bias



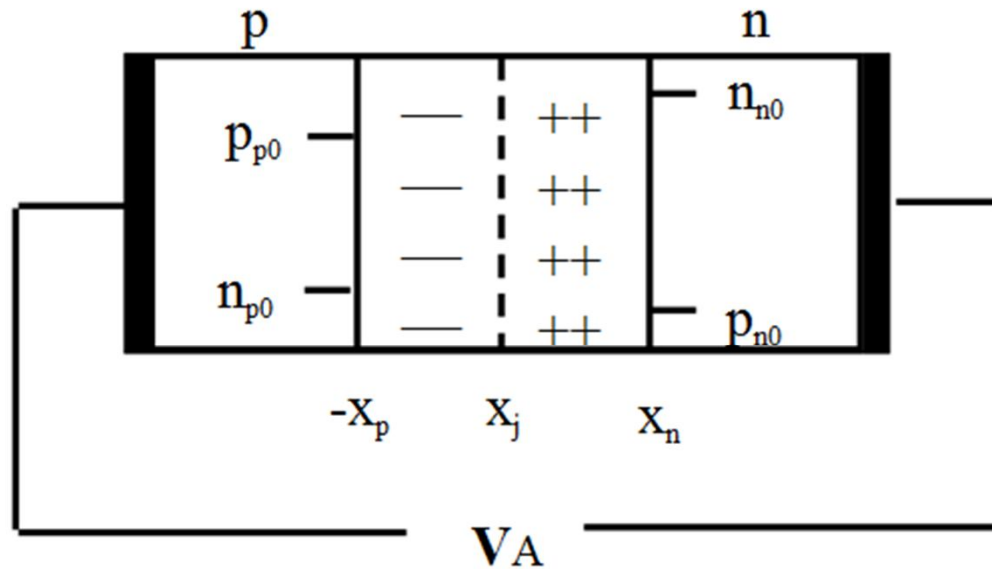
Current Voltage Characteristics

Once electrons go from $N \rightarrow P$ they become minority carriers; similarly for holes from $P \rightarrow N$. **Minority carrier behavior is of fundamental importance** in P-N junctions. The injection of minority carriers creates a concentration gradient near the edge of the depletion region. The minority carriers will therefore diffuse away from the depletion region (note: there is no electric field in the quasi-neutral region), and they will recombine if given sufficient time/distance.

The first thing needed is to relate the number of injected carriers to V_A .



Junction Definitions



Define:

n_{no} = #electrons in N region in thermal equilibrium

n_{po} = #electrons in P region in thermal equilibrium

p_{no} = #holes in N region in thermal equilibrium

p_{po} = #holes in P region in thermal equilibrium

Carrier Distribution at Zero Bias

$$n_{no} \approx N_D \text{ and } p_{no} = \frac{n_i^2}{N_D}$$

$$p_{po} \approx N_A \text{ and } n_{po} = \frac{n_i^2}{N_A}$$

$$V_{bi} = \frac{kT}{q} \ln \frac{N_D N_A}{n_i^2} = \frac{kT}{q} \ln \frac{n_{no}}{n_{po}}$$

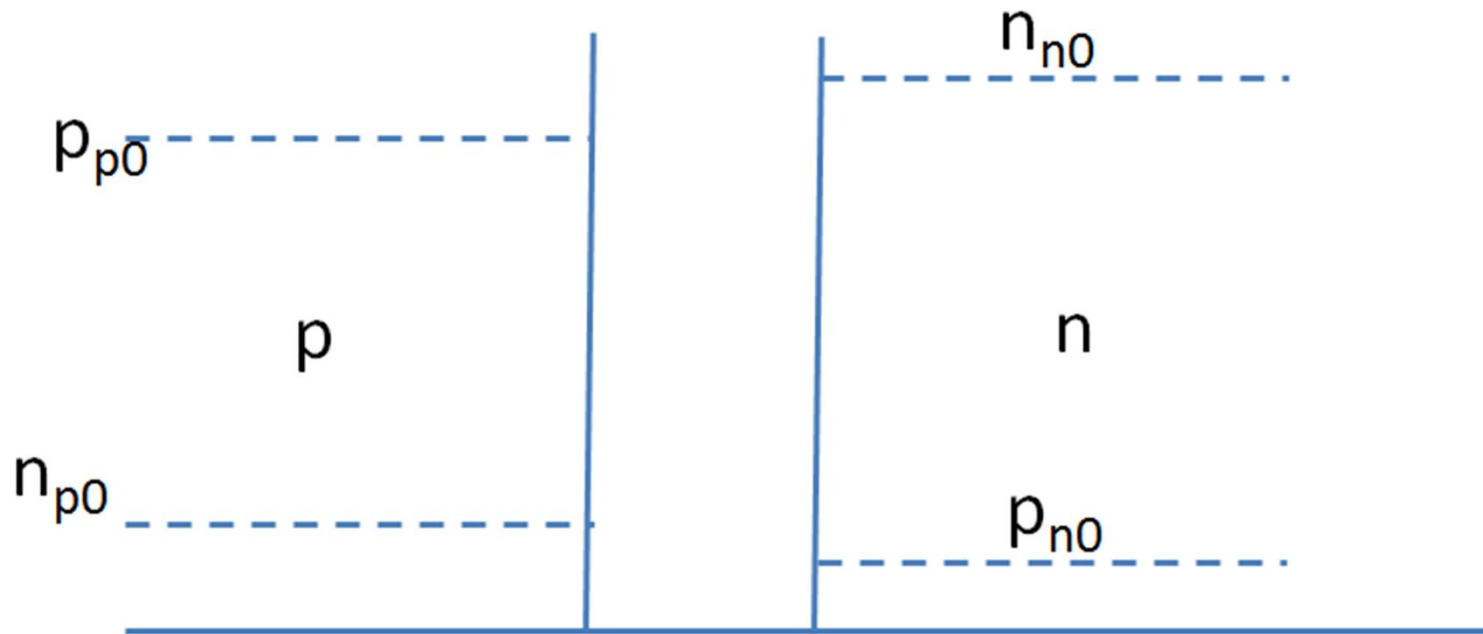
$$n_{no} = n_{po} e^{qV_{bi}/kT}$$

$$\text{similarly, } p_{po} = p_{no} e^{qV_{bi}/kT}$$

The minority carrier concentrations across the depletion region are related to the majority carrier concentrations on the other side by V_{bi} .

Complete donor and acceptor ionization

Carrier Distribution at Zero Bias



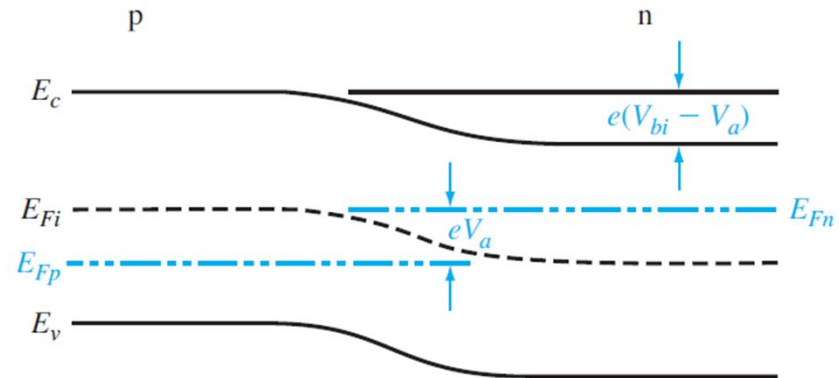
$$n_{p0} = n_{n0} \exp\left(\frac{-eV_{bi}}{kT}\right)$$

$$p_{n0} = p_{p0} \exp\left(\frac{-eV_{bi}}{kT}\right)$$

The equation relates the minority carrier electron concentration on the p-side to the majority carrier electron concentration on the n-side of the junction.

Minority Carrier Distribution under forward bias

$$n_{p0} = n_{n0} \exp\left(\frac{-eV_{bi}}{kT}\right)$$



$$n_p = n_{n0} \exp\left(\frac{-e(V_{bi} - V_a)}{kT}\right) = n_{n0} \exp\left(\frac{-eV_{bi}}{kT}\right) \exp\left(\frac{+eV_a}{kT}\right)$$

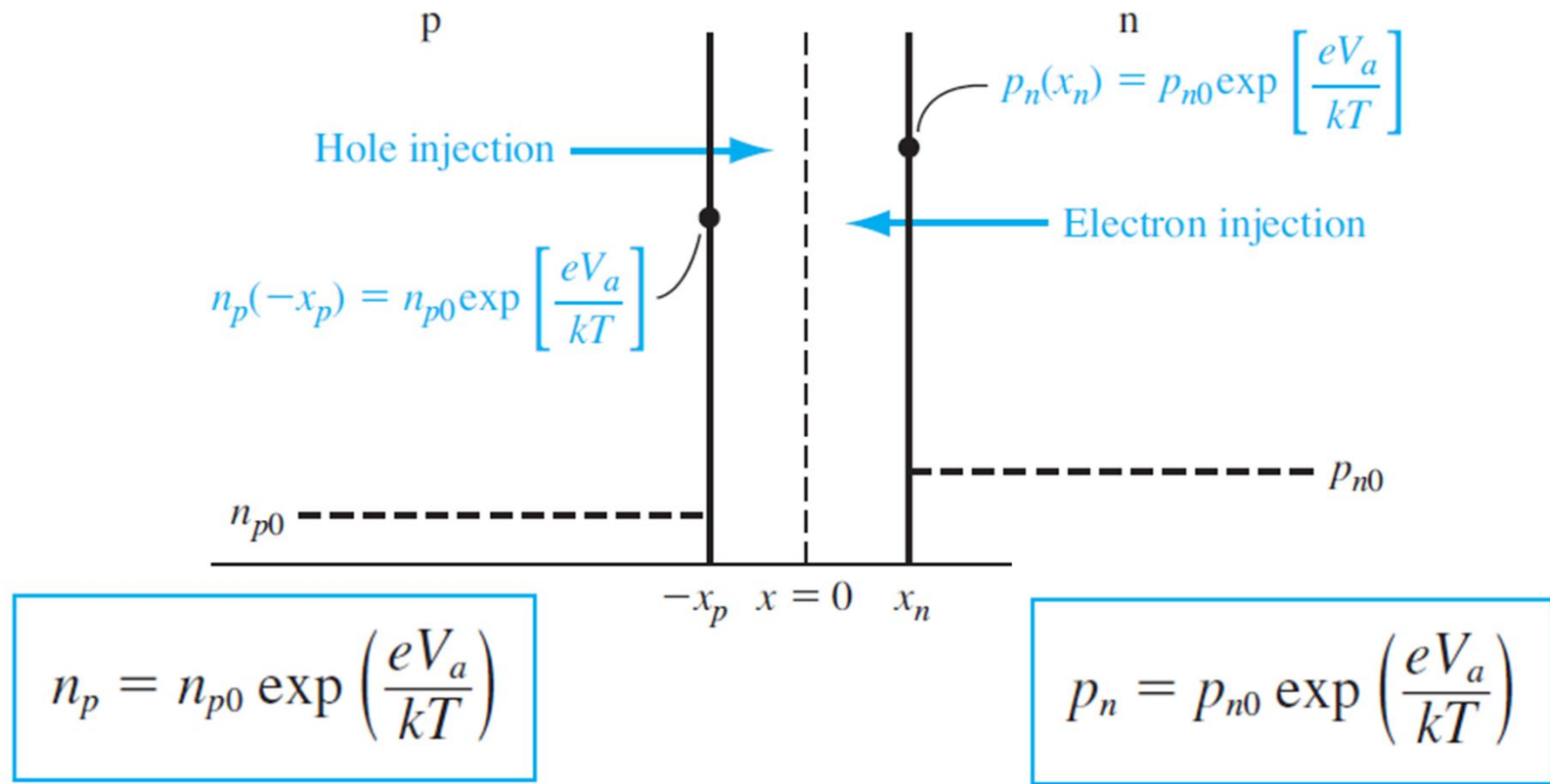
$$n_p = n_{p0} \exp\left(\frac{eV_a}{kT}\right)$$

$$p_n = p_{n0} \exp\left(\frac{eV_a}{kT}\right)$$

The minority carrier electron concentration in the p-region becomes larger than its equilibrium value when a forward bias is applied.

The same derivation can be applied for the minority carrier holes in the n-region.

Minority Carrier Distribution under forward bias



Excess carriers are subjected to the diffusion and recombination processes. The minority carrier electron concentration decreases into the p-region and the minority carrier hole concentration decreases into the n-region.

Minority Carrier Distribution under forward bias

Example: consider a silicon pn junction at $T = 300$ K with $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$. Assume $N_d = 1 \times 10^{16} \text{ cm}^{-3}$ and $V_a = 0.60$ V. Calculate the minority carrier hole concentration at the edge of the space charge region.

$$p_{n0} = \frac{n_i^2}{N_d} = \frac{(1.5 \times 10^{10})^2}{10^{16}} = 2.25 \times 10^4 \text{ cm}^{-3}$$

$$p_n = p_{n0} \exp\left(\frac{eV_a}{kT}\right) = (2.25 \times 10^4) \exp\left(\frac{0.60}{0.0259}\right) = 2.59 \times 10^{14} \text{ cm}^{-3}$$

The minority carrier hole concentration increases by many orders of magnitude.

Minority Carrier Distribution under forward bias

The ambipolar transport equation describes the distribution of the excess minority carriers as a function of time and spatial coordinates. For the minority carrier holes in the n-region, we have

$$D_p \frac{\partial^2(\delta p_n)}{\partial x^2} - \mu_p E \frac{\partial(\delta p_n)}{\partial x} + g' - \frac{\delta p_n}{\tau_{p0}} = \frac{\partial(\delta p_n)}{\partial t}$$

$$\delta p_n = p_n - p_{n0} \quad g' \text{ is the carrier generation rate.}$$

We assume that the electric field is zero in both the neutral n- and p-regions. We thus have that for $x > x_n$, $E = 0$ and $g' = 0$. If we consider the steady state, so $\partial(\delta p_n)/\partial t = 0$.

$$D_p \frac{d^2(\delta p_n)}{dx^2} - \frac{\delta p_n}{\tau_{p0}} = 0 \quad (x > x_n)$$

Minority Carrier Distribution under forward bias

$$\text{Let } L_p^2 = D_p \tau_{p0}$$

$$\frac{d^2(\delta p_n)}{dx^2} - \frac{\delta p_n}{L_p^2} = 0 \quad (x > x_n)$$

For the same set of conditions:

$$L_n^2 = D_n \tau_{n0}$$

$$\frac{d^2(\delta n_p)}{dx^2} - \frac{\delta n_p}{L_n^2} = 0 \quad (x < -x_p)$$

The general solutions are:

$$\delta p_n(x) = p_n(x) - p_{n0} = A e^{x/L_p} + B e^{-x/L_p} \quad (x > x_n)$$

$$\delta n_p(x) = n_p(x) - n_{p0} = C e^{x/L_n} + D e^{-x/L_n} \quad (x < -x_p)$$

Minority Carrier Distribution under forward bias

We have boundary conditions:

$$p_n(x_n) = p_{n0} \exp\left(\frac{eV_a}{kT}\right)$$

$$p_n(x \rightarrow +\infty) = p_{n0}$$

$$n_p(-x_p) = n_{p0} \exp\left(\frac{eV_a}{kT}\right)$$

$$n_p(x \rightarrow -\infty) = n_{p0}$$

$$\Rightarrow A = 0$$

$$\Rightarrow D = 0$$

Minority Carrier Distribution under forward bias

$$p_{n0} \exp\left(\frac{eV_a}{kT}\right) - p_{n0} = B e^{-x_n/L_p}$$

$$n_{p0} \exp\left(\frac{eV_a}{kT}\right) - n_{p0} = C e^{-x_p/L_n}$$

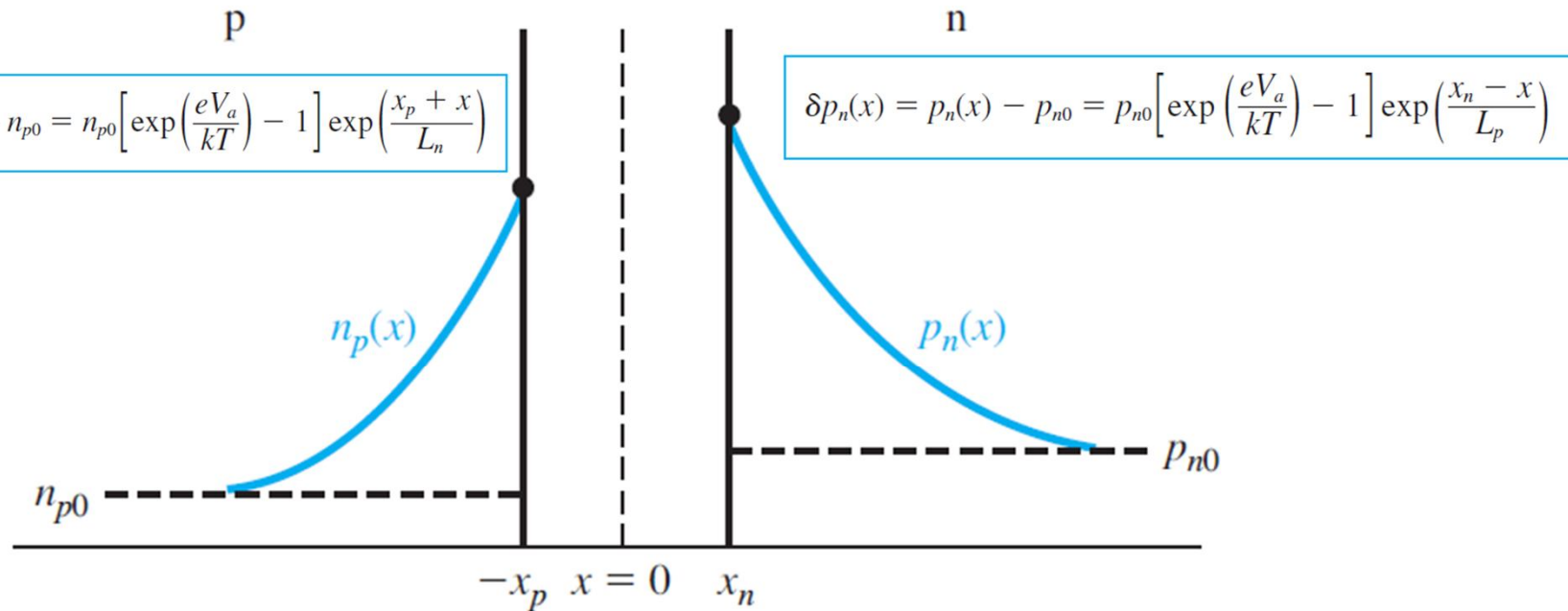
$$B = p_{n0} \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right] e^{x_n/L_p}$$

$$C = n_{p0} \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right] e^{x_p/L_n}$$

$$\delta p_n(x) = p_n(x) - p_{n0} = p_{n0} \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right] \exp\left(\frac{x_n - x}{L_p}\right) \quad (x > x_n)$$

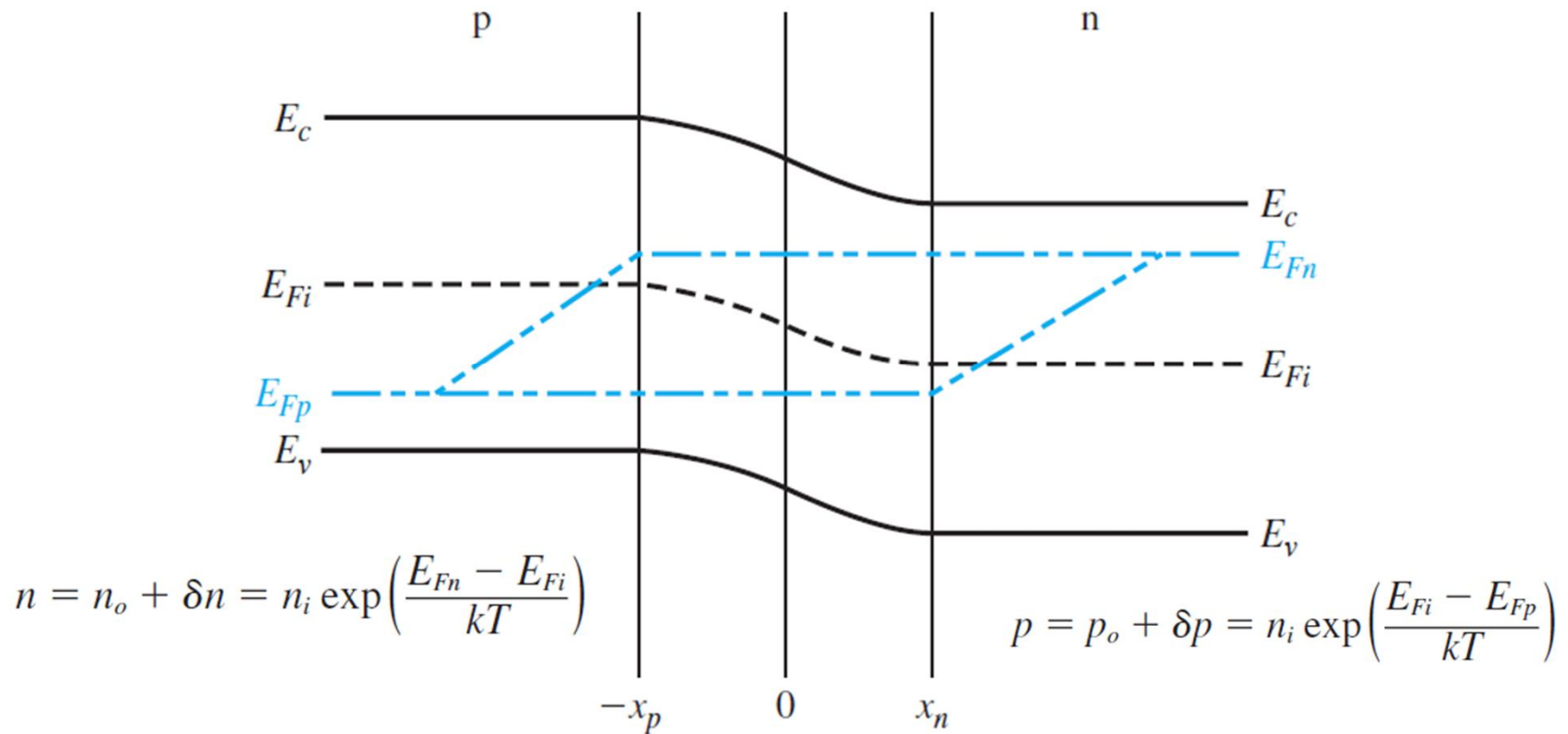
$$\delta n_p(x) = n_p(x) - n_{p0} = n_{p0} \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right] \exp\left(\frac{x_p + x}{L_n}\right) \quad (x < -x_p)$$

Minority Carrier Distribution under forward bias



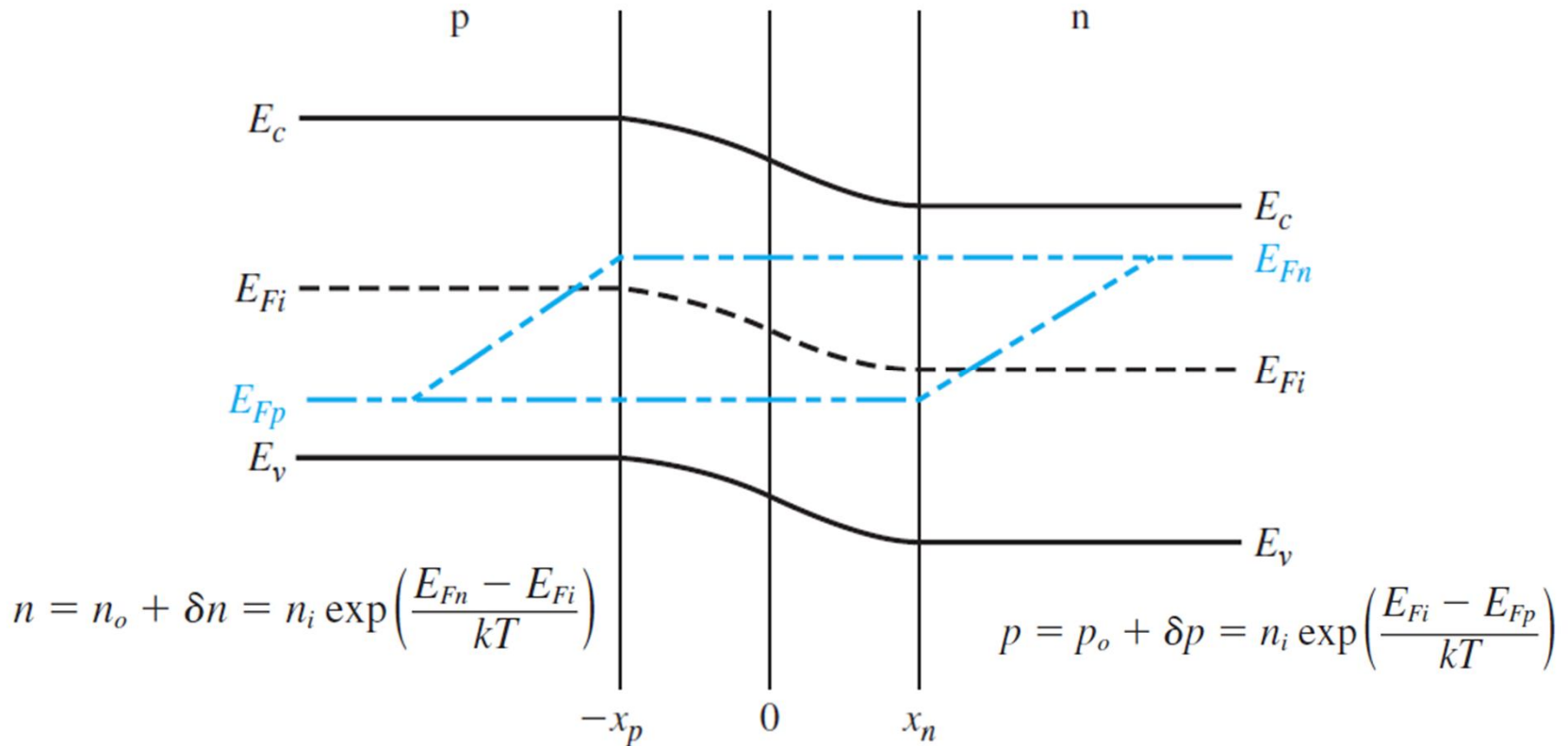
To summarize, a forward-bias voltage lowers the built-in potential barrier of a pn junction so that the electrons from the n-region are injected across the space charge region to create excess minority carriers in the p-region. These excess electrons diffuse into the bulk p-region, where they recombine with the majority carrier holes. The excess minority carrier electron concentration decreases with distance from the junction. The same discussion applies for the holes injected from the p-region into the n-region.

Quasi-Fermi levels through a forward-biased pn junction



We may note that close to the space charge edge in the p region, $E_{Fn} - E_{Fi} > 0$ which means that $\delta n > n_i$. Further from the space charge edge, $E_{Fn} - E_{Fi} < 0$ which means that $\delta n < n_i$ and the excess electron concentration is approaching zero. The same discussion applies to the excess hole concentration in the n region.

Quasi-Fermi levels through a forward-biased pn junction



$$np = n_i^2 \exp\left(\frac{E_{Fn} - E_{Fp}}{kT}\right) \quad n_o p_n(x_n) = n_o p_{no} \exp\left(\frac{V_a}{V_t}\right) = n_i^2 \exp\left(\frac{V_a}{V_t}\right)$$

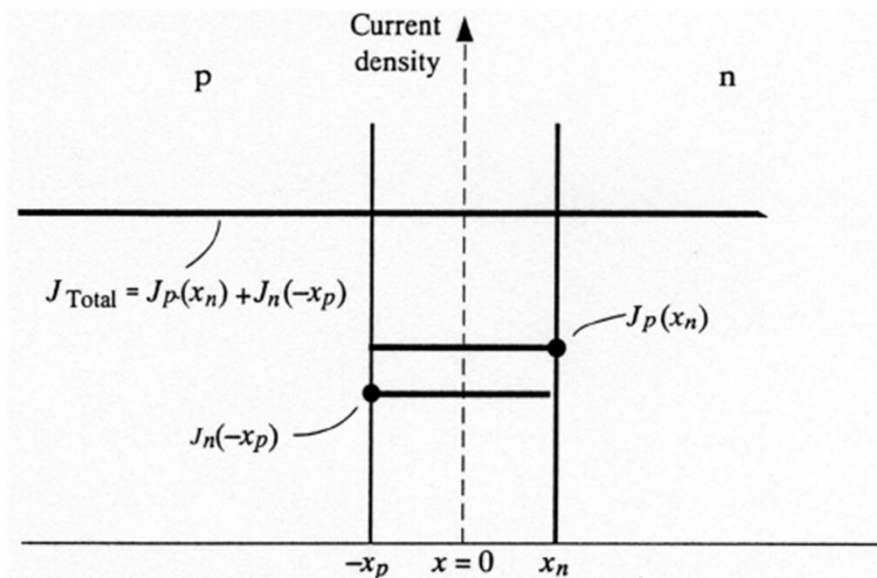
The difference in quasi-Fermi levels is related to the applied bias V_a and represents the deviation from thermal equilibrium. The difference between E_{Fn} and E_{Fp} is nearly constant through the depletion region.

Ideal Current/Voltage Relationship

- The space charge regions have abrupt boundaries and the semiconductor is neutral outside of the depletion region.
 - The Maxwell-Boltzmann approximation applies to carrier statistics.
 - Complete ionization.
 - The electric field in the bulk p- and n-regions is very small. Essentially all of the applied voltage is across the junction region.
 - The concept of low injection applies.
-
- The total current is a constant through the entire pn structure.
 - The individual electron and hole currents are continuous functions through the pn structure
 - The individual electron and hole currents are constant throughout the depletion region.

Ideal Current/Voltage Relationship

- From the assumptions, the current through the junction under a forward bias can be determined. The total current will be the sum of the electron and hole currents at any spatial location, including both drift and diffusion currents.
- Because the electron and hole currents are constant throughout the depletion region, only the hole current at $x = x_n$ and the electron current at $x = -x_p$ need to be considered.
- Because the electric field at the space charge edges is assumed zero, the hole drift current at $x = x_n$ and electron drift current at $x = -x_p$ is neglected.



$$J_p(x_n) = -eD_p \left. \frac{dp_n(x)}{dx} \right|_{x=x_n}$$

$$J_n(-x_p) = eD_n \left. \frac{dn_p(x)}{dx} \right|_{x=-x_p}$$

Ideal Current/Voltage Relationship

$$\delta p_n(x) = p_n(x) - p_{n0} = p_{n0} \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right] \exp\left(\frac{x_n - x}{L_p}\right) \quad (x > x_n)$$

$$\left. \frac{dp_n(x)}{dx} \right|_{x=x_n} = -\frac{1}{L_p} p_{n0} \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right]$$

$$J_p(x_n) = -eD_p \left. \frac{dp_n(x)}{dx} \right|_{x=x_n} = \frac{eD_p p_{n0}}{L_p} \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right]$$

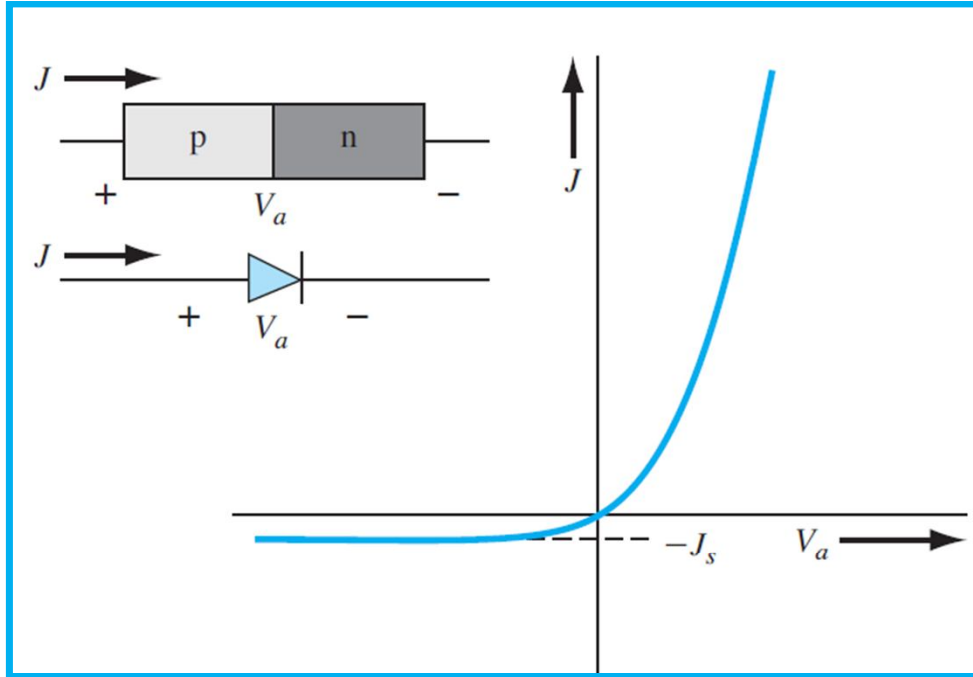
$$\delta n_p(x) = n_p(x) - n_{p0} = n_{p0} \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right] \exp\left(\frac{x_p + x}{L_n}\right) \quad (x < -x_p)$$

$$\left. \frac{dn_p(x)}{dx} \right|_{x=-x_p} = \frac{1}{L_n} n_{p0} \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right]$$

$$J_n(-x_p) = eD_n \left. \frac{dn_p(x)}{dx} \right|_{x=-x_p} = \frac{eD_n n_{p0}}{L_n} \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right]$$

Ideal Current/Voltage Relationship

$$J = J_p(x_n) + J_n(-x_p) = \left[\frac{eD_p p_{n0}}{L_p} + \frac{eD_n n_{p0}}{L_n} \right] \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right]$$



$$J_s = \frac{eD_p p_{n0}}{L_p} + \frac{eD_n n_{p0}}{L_n}$$

$$J = J_s \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right]$$

Ideal-diode equation

There is nothing to prevent V_a to be negative. If V_a becomes negative by a few kT/e volts, then the reverse-bias current density becomes independent of the reverse-bias voltage. J_s is called the **reverse-saturation current density**.

Ideal Current/Voltage Relationship

Example: consider a silicon pn junction at $T = 300$ K that is forward biased at $V_a = 0.60$ V. Consider the p-region to be doped to $N_a = 3 \times 10^{15} \text{ cm}^{-3}$ and assume the following parameters for the minority carrier electrons: $D_n = 25 \text{ cm}^2/\text{s}$, $\tau_{n0} = 10^{-7}$ s. Determine the electron diffusion current density at the edge of the space charge region.

$$J_n(-x_p) = \frac{eD_n n_{p0}}{L_n} \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right] = e \sqrt{\frac{D_n}{\tau_{n0}}} \frac{n_i^2}{N_a} \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right]$$

$$J_n(-x_p) = (1.6 \times 10^{-19}) \sqrt{\frac{25}{10^{-7}}} \frac{(1.5 \times 10^{10})^2}{3 \times 10^{15}} \left[\exp\left(\frac{0.60}{0.0259}\right) - 1 \right] = 2.18 \text{ A/cm}^2$$

Ideal Current/Voltage Relationship

Example: consider the following parameters for a silicon pn junction:

$$N_a = N_d = 10^{16} \text{ cm}^{-3},$$

$$n_i = 1.5 \times 10^{10} \text{ cm}^{-3},$$

$$D_n = 25 \text{ cm}^2/\text{sec},$$

$$\tau_{p0} = \tau_{n0} = 5 \times 10^{-7} \text{ s},$$

$$D_p = 10 \text{ cm}^2/\text{sec},$$

$$\epsilon_r = 11.7.$$

Determine the ideal reverse-saturation current density.

$$J_s = \frac{eD_n n_{p0}}{L_n} + \frac{eD_p p_{n0}}{L_p} = \frac{eD_n n_i^2}{N_a \sqrt{D_n \tau_{n0}}} + \frac{eD_p n_i^2}{N_d \sqrt{D_p \tau_{p0}}}$$

$$J_s = en_i^2 \left[\frac{1}{N_a} \sqrt{\frac{D_n}{\tau_{n0}}} + \frac{1}{N_d} \sqrt{\frac{D_p}{\tau_{p0}}} \right]$$

$$J_s = (1.6 \times 10^{-19}) (1.5 \times 10^{10})^2 \left[\frac{1}{10^{16}} \sqrt{\frac{25}{5 \times 10^{-7}}} + \frac{1}{10^{16}} \sqrt{\frac{10}{5 \times 10^{-7}}} \right] = 4.16 \times 10^{-11} \text{ A/cm}^2$$

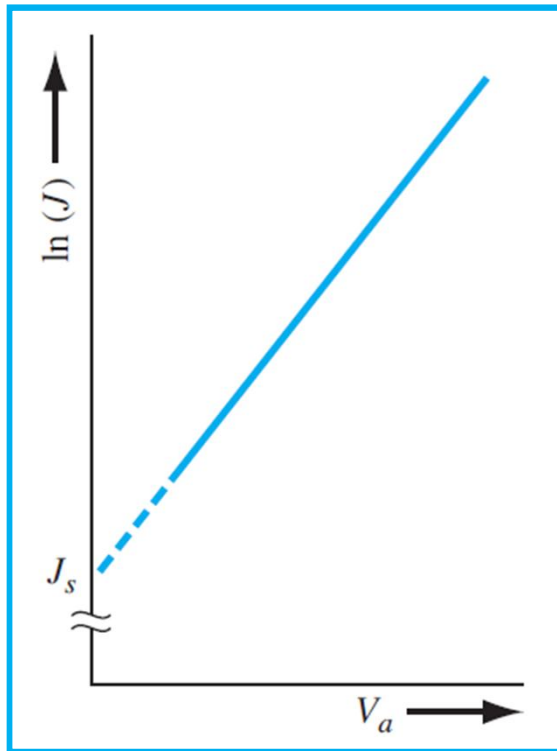
The ideal reverse-bias saturation current density is very small.

Ideal Current/Voltage Relationship

$$J = J_s \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right]$$

At $T = 300$ K, $kT/e = 0.0259$ V.

When $V_a = 5(kT/e) = 0.130$ V, $\exp(eV_a/kT) = 148 \gg 1$.



Ideally, this plot yields a straight line when V_a is greater than a few kT/e V. The forward-bias current is an exponential function of the forward-bias voltage.

Ideal Current/Voltage Relationship

For $x > x_n$:

$$\delta p_n(x) = p_n(x) - p_{n0} = p_{n0} \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right] \exp\left(\frac{x_n - x}{L_p}\right)$$

$$J_p(x) = -eD_p \frac{dp_n(x)}{dx} = \frac{eD_p p_{n0}}{L_p} \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right] \exp\left(\frac{x_n - x}{L_p}\right)$$

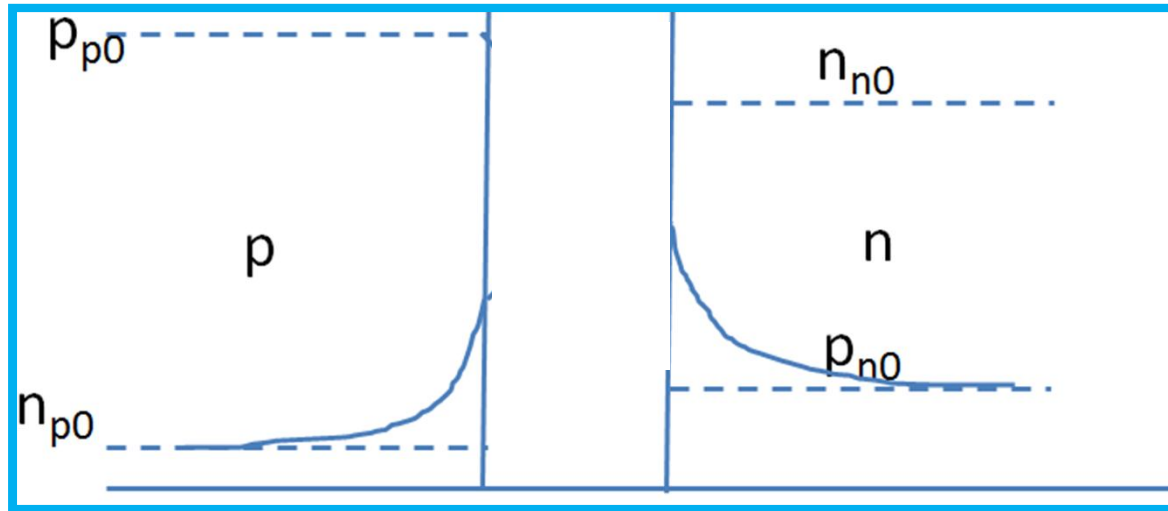
For $x < -x_p$:

$$\delta n_p(x) = n_p(x) - n_{p0} = n_{p0} \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right] \exp\left(\frac{x_p + x}{L_n}\right)$$

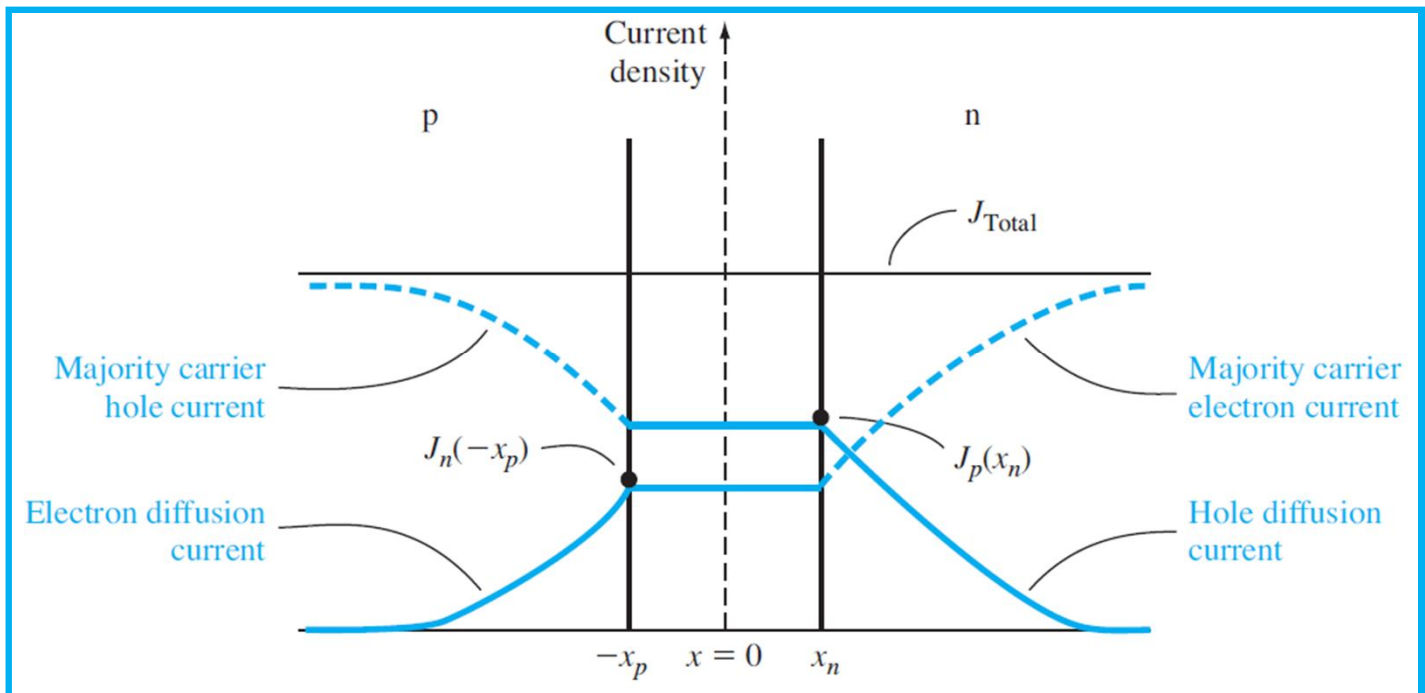
$$J_n(x) = eD_n \frac{dn_p(x)}{dx} = \frac{eD_n n_{p0}}{L_n} \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right] \exp\left(\frac{x_p + x}{L_n}\right)$$

Ideal Current/Voltage Relationship

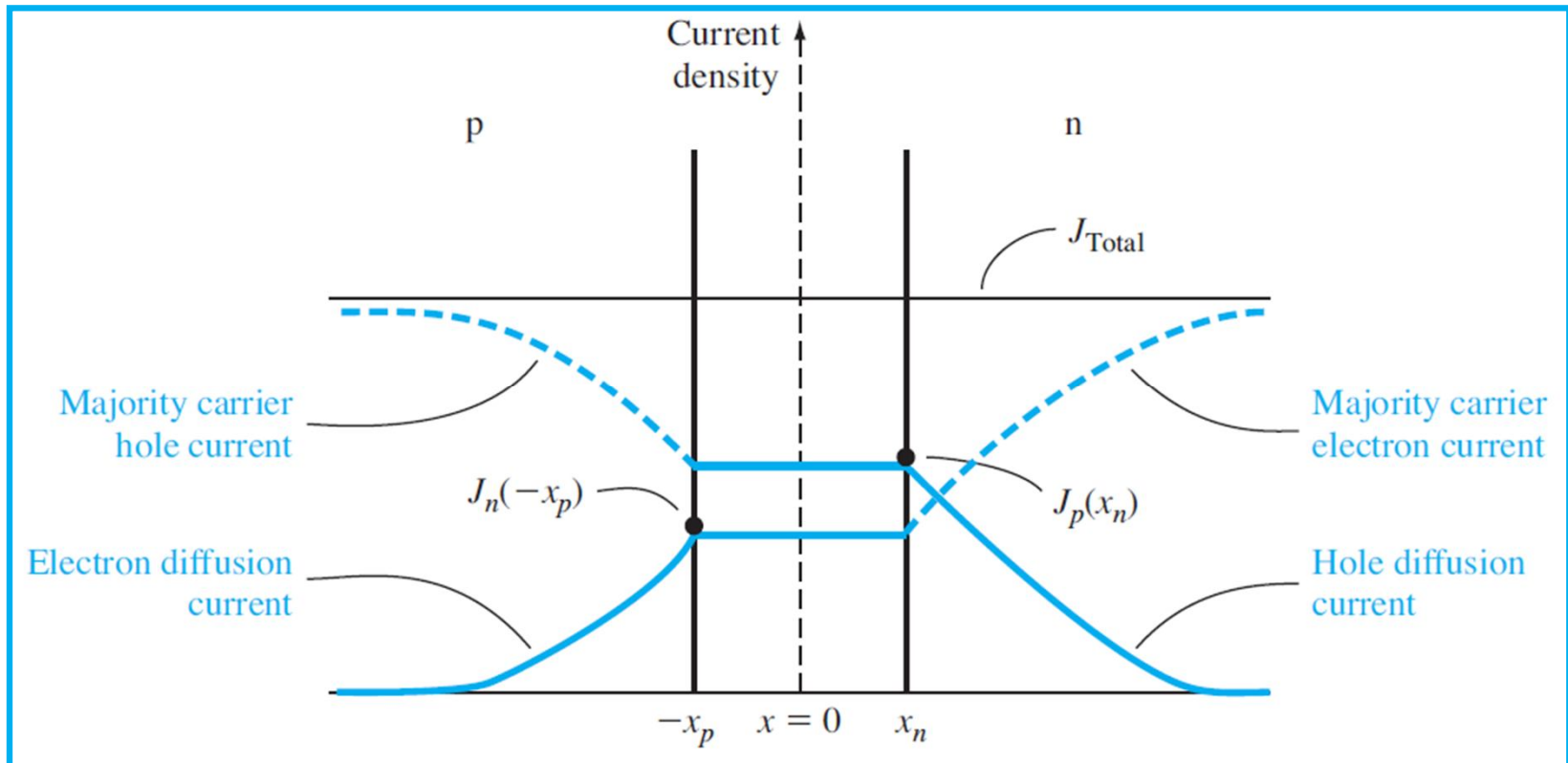
**Carrier
distribution
under
forward
bias**



**Current
distribution
under
forward
bias**

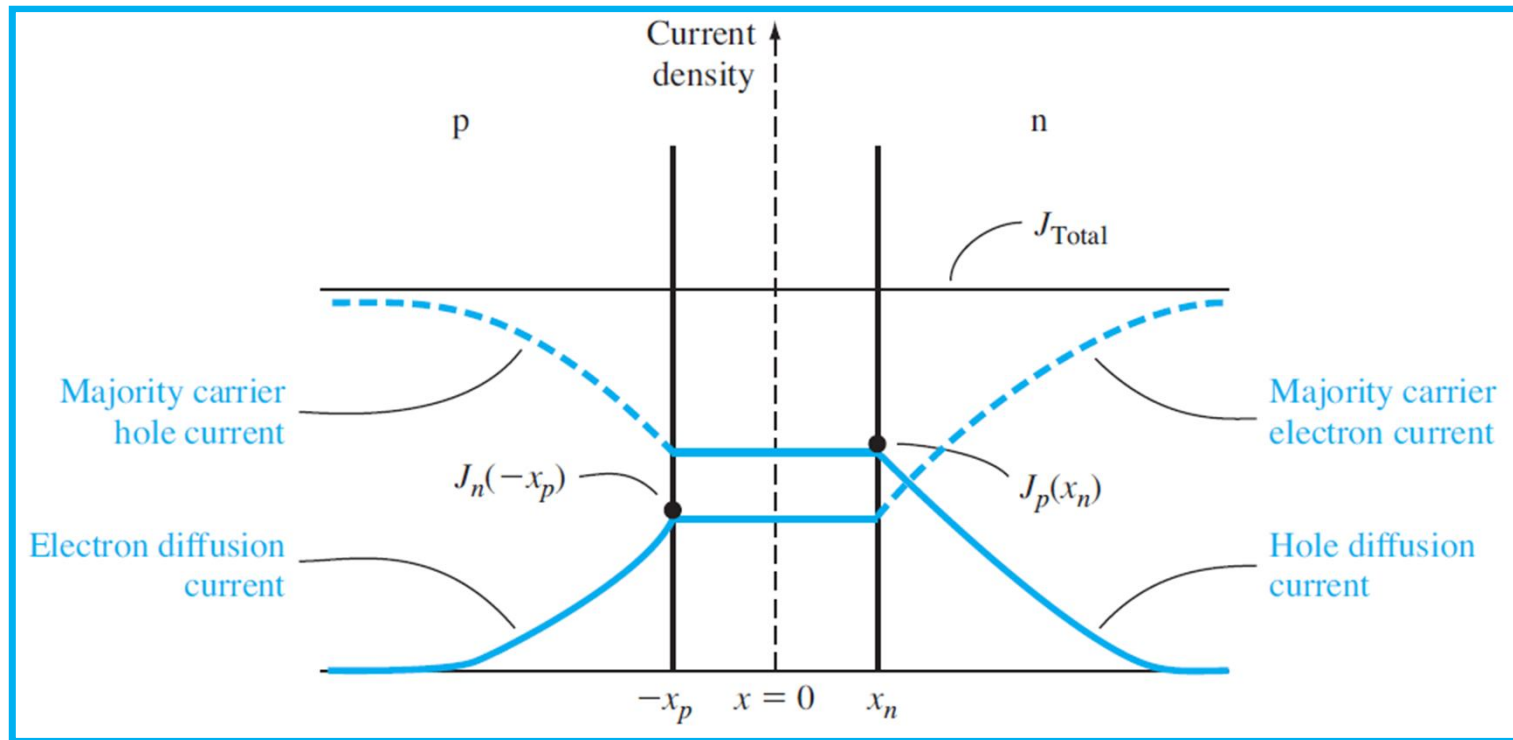


Ideal Current/Voltage Relationship



The minority carrier diffusion current density decreases with distance away from the junction. Since the total current density is constant, the majority carrier current density must increase. Because the majority carrier concentrations are constant, the majority carrier current is due to the carrier drift. The drift of the majority carriers towards the junction is to supply the carriers that are being injected across the space charge region and also to supply the carriers that are lost by recombination with the excess minority carriers.

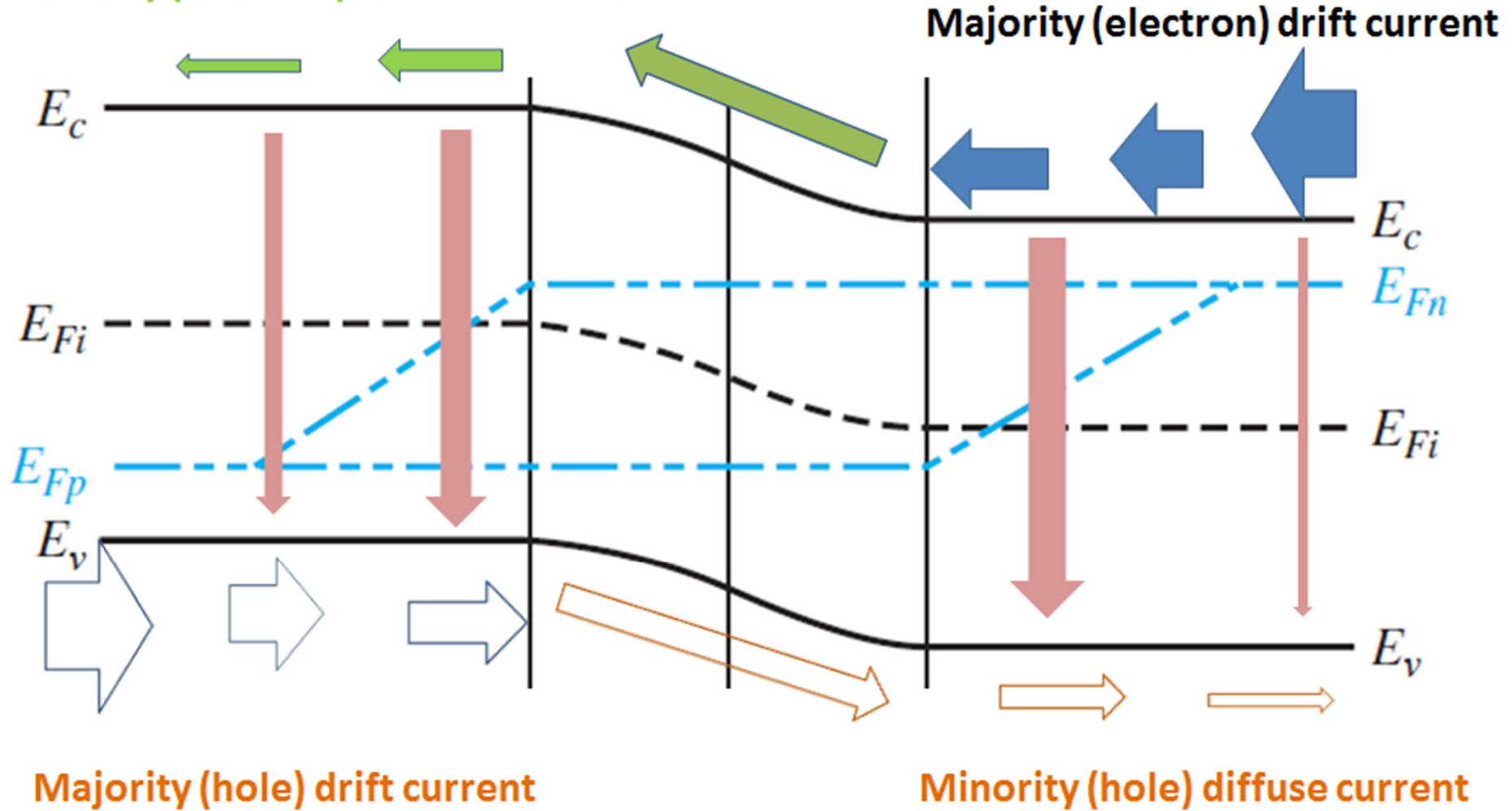
Ideal Current/Voltage Relationship



As minority carriers recombine, current is transferred to majority carriers. The resulting total current is constant throughout the device. A key point is that there is no recombination in the depletion region, so the **electron and hole currents are constant** across the region. This is the same assumption made in deriving the law of the junction. We can therefore apply our boundary conditions at the edge of the depletion region and **solve for each carrier separately**.

Ideal Current/Voltage Relationship

Minority (electron) diffuse current



Ideal Current/Voltage Relationship

minority carriers move primarily by diffusion, while the majority carriers are pulled to the junction region by drift (E).

The minority carriers control the behavior of the P-N junctions. This is in contrast to Schottky diodes (coming later) where majority carriers dominate.

$$J_s = \frac{eD_p p_{n0}}{L_p} + \frac{eD_n n_{p0}}{L_n}$$

This last result illustrates particularly well that it is the doping level on the **LIGHTLY DOPED** side of the junction that determines the current. In other words, majority carriers from the heavily doped side of the junction are injected as minority carriers into the lightly doped side. The resulting concentration gradient and greater number of minority carriers on the lightly doped side primarily determine the magnitude of the current.

Ideal Current/Voltage Relationship

Example: consider a silicon pn junction at $T = 300$ K with the following parameters and an applied forward-bias voltage $V_a = 0.65$ V:

$$\begin{aligned}N_a &= N_d = 10^{16} \text{ cm}^{-3}, & n_i &= 1.5 \times 10^{10} \text{ cm}^{-3}, \\D_n &= 25 \text{ cm}^2/\text{s}, & \tau_{p0} &= \tau_{n0} = 5 \times 10^{-7} \text{ s}, \\D_p &= 10 \text{ cm}^2/\text{s}, & \epsilon_r &= 11.7, & \mu_n &= 1350 \text{ cm}^2/(\text{V}\cdot\text{s}).\end{aligned}$$

Determine the electric field required to produce the majority carrier electron drift current far away from the junction in the n-region.

$$J_s = en_i^2 \left[\frac{1}{N_a} \sqrt{\frac{D_n}{\tau_{n0}}} + \frac{1}{N_d} \sqrt{\frac{D_p}{\tau_{p0}}} \right] = 4.16 \times 10^{-11} \text{ A/cm}^2$$

$$J = (4.16 \times 10^{-11}) \left[\exp\left(\frac{0.65}{0.0259}\right) - 1 \right] = 3.30 \text{ A/cm}^2$$

$$J_n \cong e\mu_n N_d E$$

$$E = \frac{J_n}{e\mu_n N_d} = \frac{3.30}{(1.6 \times 10^{-19})(1350)(10^{16})} = 1.53 \text{ V/cm}$$

The electric field in the neutral region is very small. Therefore the approximation of zero electric field in the bulk p- and n-regions is very good.

Ideal Current/Voltage Relationship

$$N_c = 2 \left(\frac{2\pi m_n^* kT}{h^2} \right)^{3/2}$$

$$N_v = 2 \left(\frac{2\pi m_p^* kT}{h^2} \right)^{3/2}$$

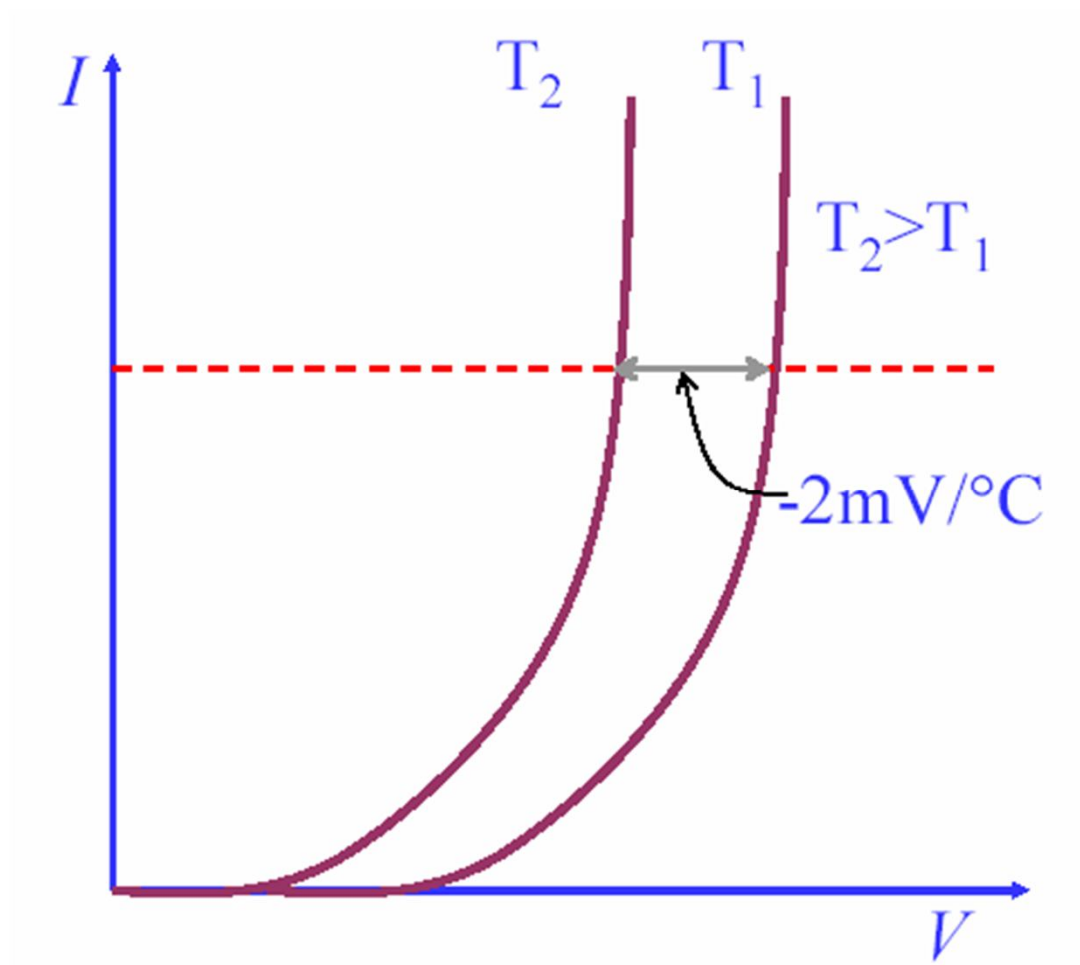
$$n_i^2 = N_c N_v \exp \left[\frac{-E_g}{kT} \right]$$

$$J_s = \frac{eD_n n_{p0}}{L_n} + \frac{eD_p p_{n0}}{L_p} = \frac{eD_n n_i^2}{N_a \sqrt{D_n \tau_{n0}}} + \frac{eD_p n_i^2}{N_d \sqrt{D_p \tau_{p0}}}$$
$$J_s = en_i^2 \left[\frac{1}{N_a} \sqrt{\frac{D_n}{\tau_{n0}}} + \frac{1}{N_d} \sqrt{\frac{D_p}{\tau_{p0}}} \right] \propto (kT)^3 \exp \left(\frac{-E_g}{kT} \right)$$

$$J = J_s \left[\exp \left(\frac{eV_a}{kT} \right) - 1 \right] \propto (kT)^3 \exp \left(\frac{-E_g}{kT} \right) \exp \left(\frac{eV_a}{kT} \right)$$

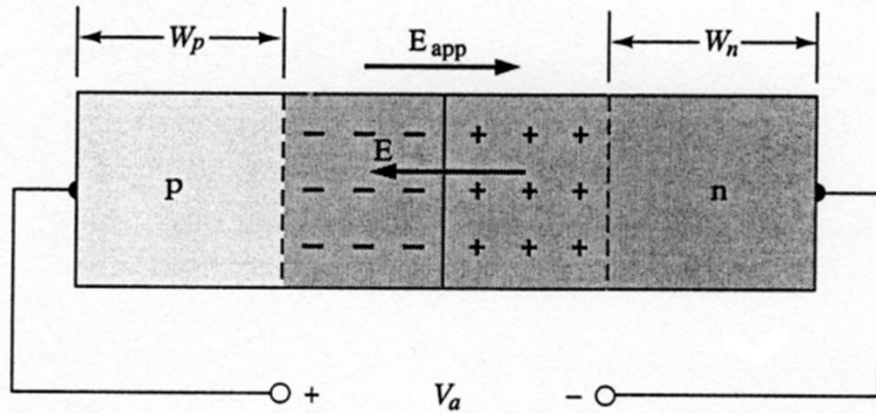
The forward-bias current is a strong function of temperature.

Temperature Effects



Diode current will increase as a specific voltage for both bias polarities

The “long” Diode



It means that the lengths of both neutral regions, W_p and W_n , are very long. The characteristic length scales for comparison are the minority carrier diffusion lengths, L_p and L_n .

$$\frac{d^2(\delta p_n)}{dx^2} - \frac{\delta p_n}{L_p^2} = 0 \quad (x > x_n)$$

$$\frac{d^2(\delta n_p)}{dx^2} - \frac{\delta n_p}{L_n^2} = 0 \quad (x < -x_p)$$

The boundary conditions:

$$p_n(x_n) = p_{n0} \exp\left(\frac{eV_a}{kT}\right)$$

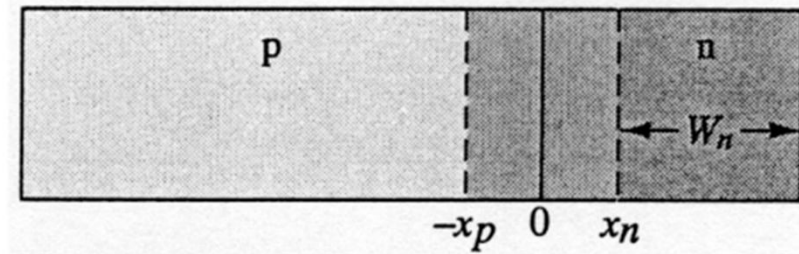
$$p_n(x \rightarrow +\infty) = p_{n0}$$

$$n_p(-x_p) = n_{p0} \exp\left(\frac{eV_a}{kT}\right)$$

$$n_p(x \rightarrow -\infty) = n_{p0}$$

The “Short” Diode

In many pn junction structures, one region may be short compared to the minority carrier diffusion length. For example, $W_n \ll L_p$.



$$\frac{d^2(\delta p_n)}{dx^2} - \frac{\delta p_n}{L_p^2} = 0 \quad (x > x_n)$$

$$p_n(x_n) = p_{n0} \exp\left(\frac{eV_a}{kT}\right)$$

If we assume an ohmic contact at $x = (x_n + W_n)$, the surface-recombination velocity will be infinitely large and the excess minority carrier concentration will be zero:

$$p_n(x = x_n + W_n) = p_{n0}$$

The “Short” Diode

The general solutions are:

$$\delta p_n(x) = p_n(x) - p_{n0} = Ae^{x/L_p} + Be^{-x/L_p} \quad (x > x_n)$$

$$\begin{cases} Ae^{x_n/L_p} + Be^{-x_n/L_p} = p_{n0} \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right] \\ Ae^{(x_n+W_n)/L_p} + Be^{-(x_n+W_n)/L_p} = 0 \end{cases}$$
$$\delta p_n(x) = p_{n0} \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right] \frac{\sinh[(x_n + W_n - x)/L_p]}{\sinh[W_n/L_p]}$$

When $W_n \gg L_p$:

$$\sinh[W_n/L_p] = \frac{1}{2} (e^{W_n/L_p} - e^{-W_n/L_p}) \approx \frac{1}{2} e^{W_n/L_p}$$

$$\sinh[(x_n + W_n - x)/L_p] \approx \frac{1}{2} e^{(x_n + W_n - x)/L_p}$$

The “Short” Diode

$$\delta p_n(x) = p_{n0} \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right] \exp\left(\frac{x_n - x}{L_p}\right)$$

When $W_n \ll L_p$:

$$\sinh[W_n / L_p] \approx \frac{W_n}{L_p}$$

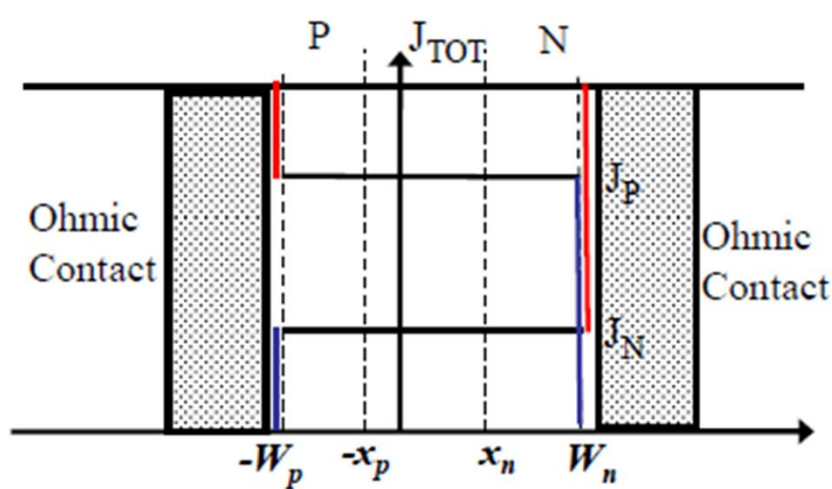
$$\sinh[(x_n + W_n - x) / L_p] \approx \frac{x_n + W_n - x}{L_p}$$

$$\delta p_n(x) = p_{n0} \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right] \left[\frac{x_n + W_n - x}{W_n} \right]$$

$$J_p(x) = -eD_p \frac{d(\delta p_n(x))}{dx} = \frac{eD_p p_{n0}}{W_n} \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right]$$

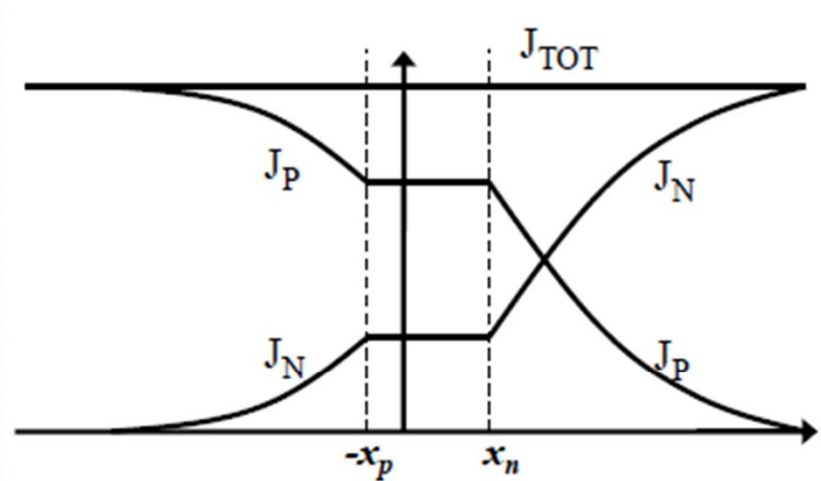
The minority carrier diffusion current is a constant, and it is larger for a short diode than for a long diode if $W_n \ll L_p$.

Comparison of short vs long base diode



$$I = qAn_i^2 \left[\frac{D_p}{N_D W_n} + \frac{D_N}{N_A W_p} \right] (e^{qV_A/kT} - 1)$$

No recombination in quasi-neutral region (J_N & J_p constant)



$$I = qAn_i^2 \left(\frac{D_p}{N_D L_p} + \frac{D_N}{N_A L_N} \right) (e^{qV_A/kT} - 1)$$

Recombination in quasi-neutral region to reach p_{n0} and n_{p0}

Only difference is diode length rather than diffusion length. The diodes are indistinguishable and both can be represented by the expression,

$$I = I_o (e^{qV_A/kT} - 1)$$

8.2 | GENERATION-RECOMBINATION CURRENTS

Number of active traps increases with reverse bias

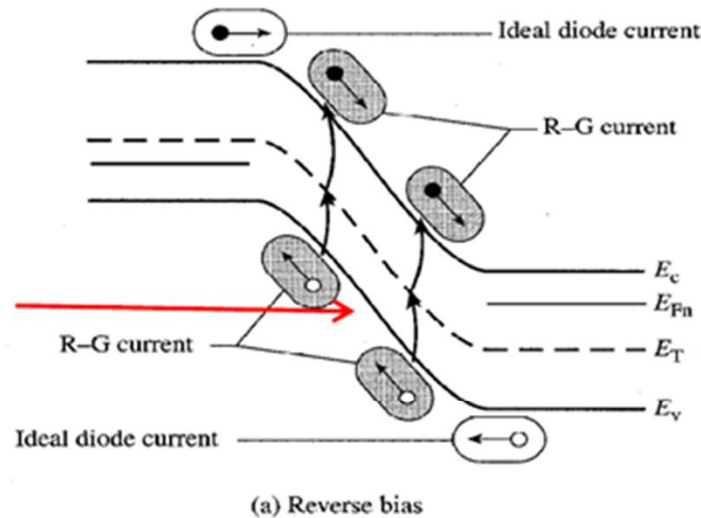
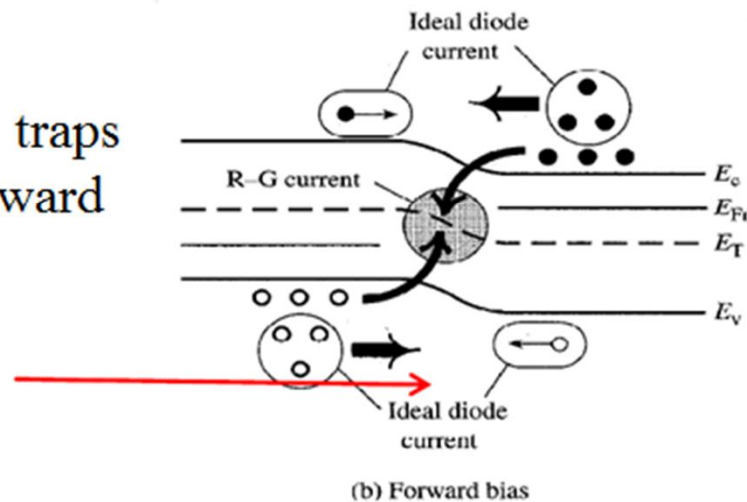


Photo detector operates in reverse bias: generation current

Number of active traps saturates with forward bias



Generation and Recombination act as **added parallel paths** to the ideal diode current

8.2 | GENERATION–RECOMBINATION CURRENTS

The actual I – V characteristics of a pn junction diode deviate from the ideal expression. There are additional current components that result from the **carrier generation and recombination** processes.

The recombination rate of excess electrons and holes, derived from the Shockley-Read-Hall recombination theory, is given by

$$R = \frac{C_n C_p N_t (np - n_i^2)}{C_n (n + n') + C_p (p + p')} \quad \begin{aligned} n' &= N_c \exp\left[\frac{-(E_c - E_t)}{kT}\right] \\ p' &= N_v \exp\left[\frac{-(E_t - E_v)}{kT}\right] \end{aligned}$$

C_n : constant proportional to electron-capture cross section

C_p : constant proportional to hole-capture cross section

N_t : total concentration of trapping centers

E_t : trap energy within the bandgap

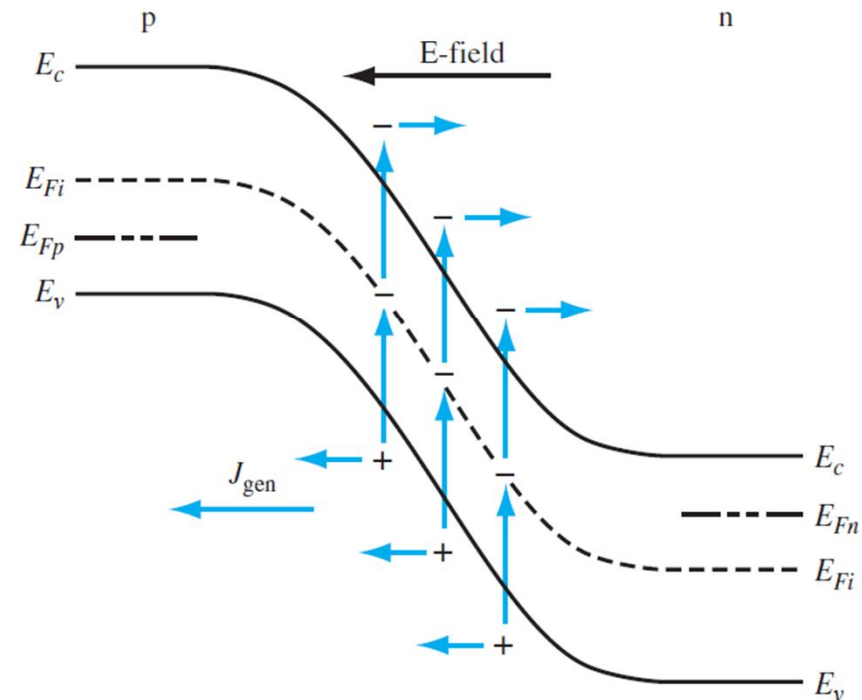
Reverse-Bias Generation Current

For a pn junction under reverse bias, mobile electrons and holes are essentially swept out of the space charge region. Accordingly, within the space charge region, $n \cong p \cong 0$. The recombination rate becomes

$$R = \frac{-C_n C_p N_t n_i^2}{C_n n' + C_p p'}$$

The negative sign implies that electron-hole pairs are really generated within the reverse-biased space charge region.

The flow of charge is in the direction of a reverse-bias current. This **reverse-bias generation current** is additional to the **ideal reverse-bias saturation current**.



Reverse-Bias Generation Current

We can calculate the reverse-bias generation current density. If we assume the trap level to be at the intrinsic Fermi level, then

$$n' = N_c \exp\left[\frac{-(E_c - E_t)}{kT}\right] = n_i \quad p' = N_v \exp\left[\frac{-(E_t - E_v)}{kT}\right] = n_i$$

$$R = \frac{-C_n C_p N_t n_i^2}{C_n n_i + C_p n_i} = \frac{-n_i}{\frac{1}{N_t C_p} + \frac{1}{N_t C_n}}$$

Because $\tau_{p0} = \frac{1}{N_t C_p}$ and $\tau_{n0} = \frac{1}{N_t C_n}$

then
$$R = \frac{-n_i}{\tau_{p0} + \tau_{n0}}$$

Reverse-Bias Generation Current

If we define a new lifetime as the average of τ_{p0} and τ_{n0} ,

$$\tau_0 = \frac{\tau_{p0} + \tau_{n0}}{2}$$

The recombination rate is then written as

$$R = \frac{-n_i}{2\tau_0} \equiv -G$$

G is the generation rate because the negative recombination rate implies a generation rate. The generation current density is the integral of the generation rate over the space charge region:

$$J_{\text{gen}} = \int_0^W eGdx = \int_0^W e \frac{n_i}{2\tau_0} dx = \frac{en_i W}{2\tau_0}$$

$$J_R = J_s + J_{\text{gen}}$$

J_s is independent of the reverse bias voltage. J_{gen} depends on the depletion width and thus is a function of the reverse-bias voltage.

Reverse-Bias Generation Current

Example: consider a silicon pn junction under reverse bias at $T = 300$ K with the following parameters:

$$N_a = N_d = 10^{16} \text{ cm}^{-3},$$

$$n_i = 1.5 \times 10^{10} \text{ cm}^{-3},$$

$$D_n = 25 \text{ cm}^2/\text{sec},$$

$$\tau_0 = \tau_{p0} = \tau_{n0} = 5 \times 10^{-7} \text{ sec},$$

$$D_p = 10 \text{ cm}^2/\text{sec},$$

$$\epsilon_r = 11.7, \quad V_{bi} + V_R = 5 \text{ V}.$$

Determine the ideal reverse saturation current density and the generation current density.

$$J_s = en_i^2 \left[\frac{1}{N_a} \sqrt{\frac{D_n}{\tau_{n0}}} + \frac{1}{N_d} \sqrt{\frac{D_p}{\tau_{p0}}} \right] = 4.16 \times 10^{-11} \text{ A/cm}^2$$

$$W = \left\{ \frac{2\epsilon_s}{e} \left(\frac{N_a + N_d}{N_a N_d} \right) (V_{bi} + V_R) \right\}^{1/2} = 1.14 \times 10^{-4} \text{ cm}$$

$$J_{\text{gen}} = \frac{en_i W}{2\tau_0} = 2.74 \times 10^{-7} \text{ A/cm}^2$$

For the silicon pn junction diode at room temperature, the generation current density is approximately four orders of magnitude larger than the ideal saturation current density.

Forward-Bias Recombination Current

Under forward bias, it is possible that some injected electrons and holes will recombine within the space charge region and not become part of the minority carrier distribution.

$$R = \frac{C_n C_p N_t (np - n_i^2)}{C_n (n + n') + C_p (p + p')}$$

Divide both nominator and denominator by $C_n C_p N_t$ and use the definitions of τ_{n0} and τ_{p0} , we obtain

$$R = \frac{np - n_i^2}{\tau_{p0}(n + n') + \tau_{n0}(p + p')}$$

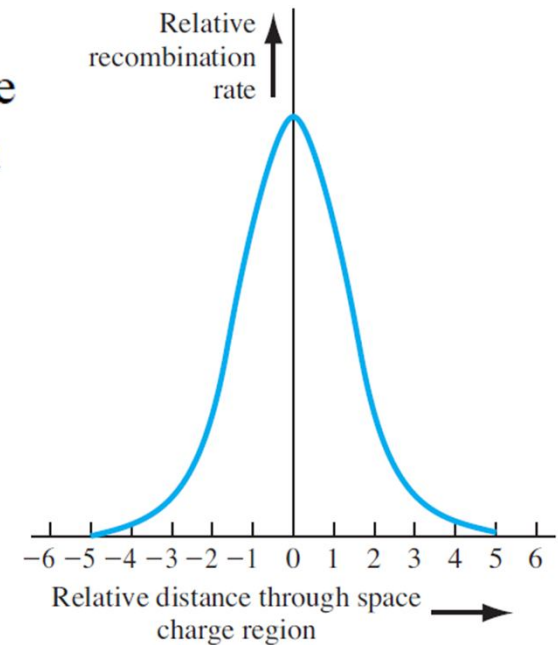
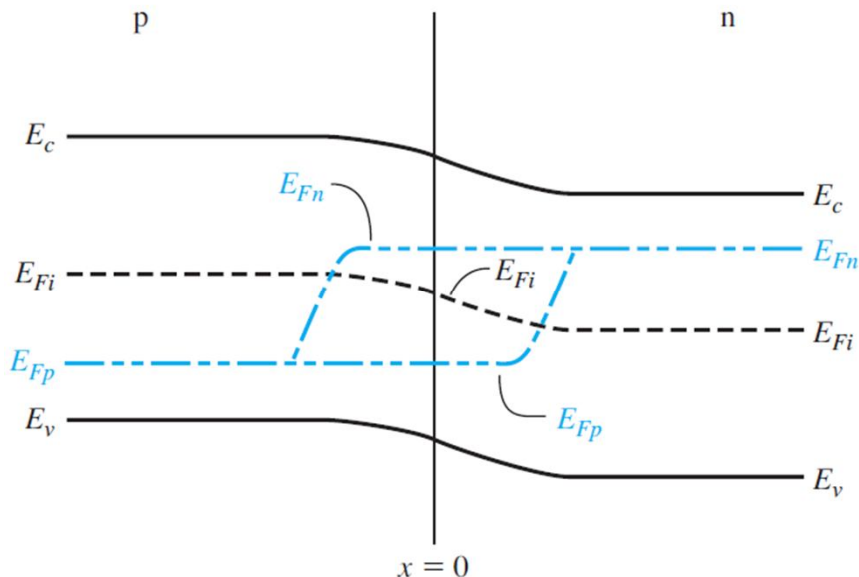
Reversely
proportional
to n+p

The electron and hole concentrations are described by:

$$n = n_i \exp\left[\frac{E_{Fn} - E_{Fi}}{kT}\right] \qquad p = n_i \exp\left[\frac{E_{Fi} - E_{Fp}}{kT}\right]$$

Forward-Bias Recombination Current

Combining the equations shown on the previous slide, we can determine the relative magnitude of the recombination rate as a function of distance through the space charge region. The maximum is at the metallurgical junction.



We can determine the maximum recombination rate. At the metallurgical junction:

$$E_{Fn} - E_{Fi} = E_{Fi} - E_{Fp} = \frac{eV_a}{2} \Rightarrow n=p$$

$$np = n_i^2 \exp\left(\frac{E_{Fn} - E_{Fp}}{kT}\right) \quad n + p \geq \sqrt{2np}, \text{ minimum value is obtained when } n=p, \text{ which sets the maximum recombination rate.}$$

Forward-Bias Recombination Current

$$n = n_i \exp\left[\frac{eV_a}{2kT}\right] \qquad p = n_i \exp\left[\frac{eV_a}{2kT}\right]$$

If we assume $n' = p' = n_i$ and $\tau_{n0} = \tau_{p0} = \tau_0$, we obtain

$$R_{\max} = \frac{n_i}{2\tau_0} \frac{[\exp(eV_a / kT) - 1]}{[\exp(eV_a / 2kT) + 1]}$$

If $V_a \gg kT/e$, the above equation becomes

$$R_{\max} = \frac{n_i}{2\tau_0} \exp\left(\frac{eV_a}{2kT}\right)$$

The recombination current density is

$$J_{\text{rec}} = \int_0^W eRdx$$

The integration is over the entire space charge region. However, the recombination rate is not a constant through the space charge region.

Forward-Bias Recombination Current

We can approximate the integration as

$$J_{\text{rec}} = ex' \frac{n_i}{2\tau_0} \exp\left(\frac{eV_a}{2kT}\right)$$

x' is a length over which the maximum recombination rate is effective. However, since τ_0 is not a well-defined or known parameter, it is customary to write the above equation as

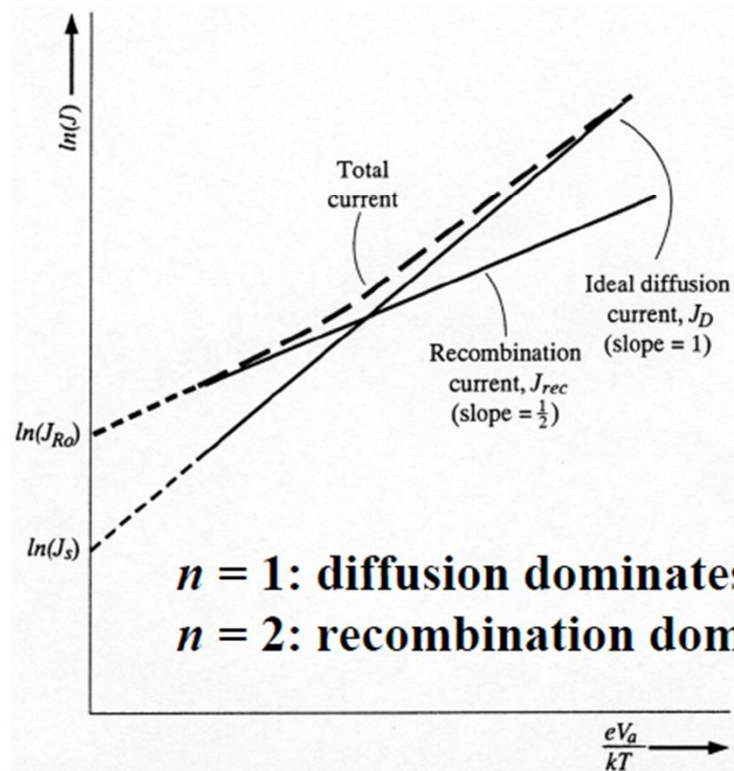
$$J_{\text{rec}} = \frac{eWn_i}{2\tau_0} \exp\left(\frac{eV_a}{2kT}\right) = J_{r0} \exp\left(\frac{eV_a}{2kT}\right)$$

The total forward-bias current density is the sum of the recombination and the ideal diffusion current densities:

$$J = J_{\text{rec}} + J_D = J_{r0} \exp\left(\frac{eV_a}{2kT}\right) + J_s \exp\left(\frac{eV_a}{kT}\right)$$

Total Forward-Bias Current

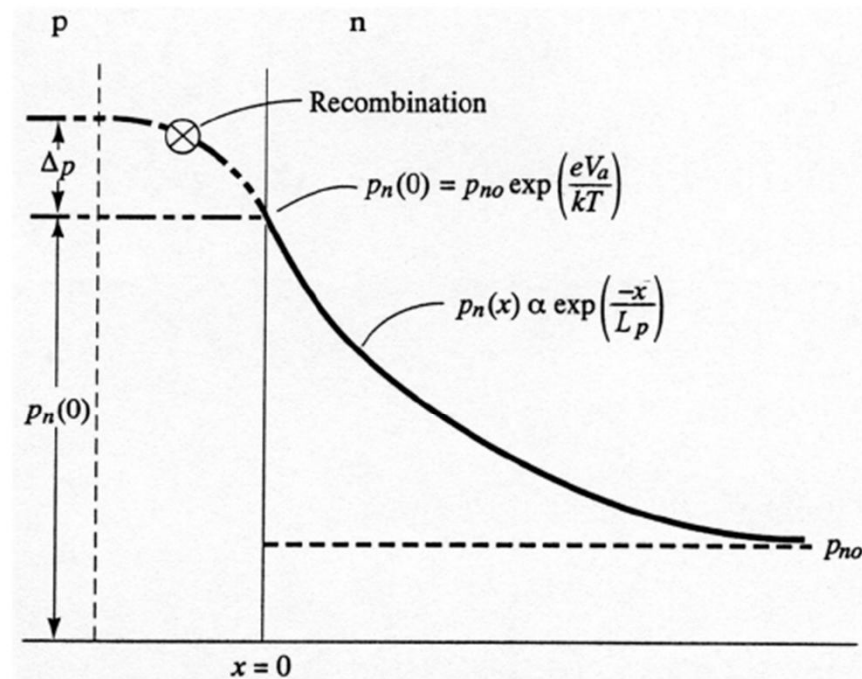
Some of the injected holes from the p-region recombine, which results in the recombination current.



$n = 1$: diffusion dominates

$n = 2$: recombination dominates

$$I = I_s \left[\exp\left(\frac{eV_a}{nkT}\right) - 1 \right]$$



Take the log of both current densities:

$$\ln J_{rec} = \ln J_{r0} + \frac{eV_a}{2kT} = \ln J_{r0} + \frac{V_a}{2V_t}$$

$$\ln J_D = \ln J_s + \frac{eV_a}{kT} = \ln J_s + \frac{V_a}{V_t}$$

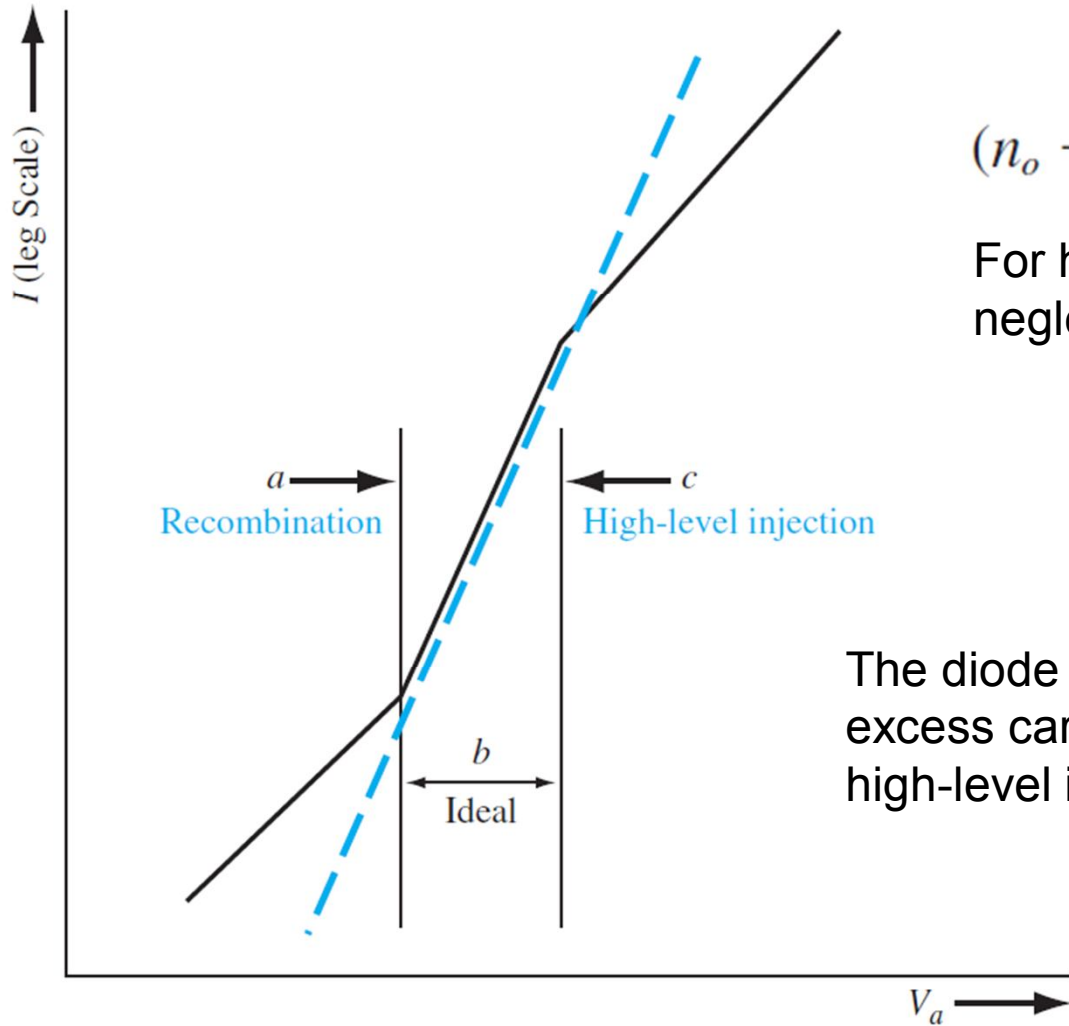
where **n** is called the **diode ideality factor**,

High level injection

We have so far assumed that the injection density of minority carriers was low so that the voltage all dropped across the depletion region. However, as the **forward bias is increased**, the injection level increases and eventually the **injected minority carrier density becomes comparable to the majority carrier density**.

When this happens, an increasingly larger fraction of the external bias drops across the undepleted region. The diode current will then saturate. The minority carriers transport is not only due to diffusion, but also due to the electric field that is now present in the **undepleted region**.

High level injection



$$np = n_i^2 \exp\left(\frac{V_a}{V_t}\right)$$

$$(n_o + \delta n)(p_o + \delta p) = n_i^2 \exp\left(\frac{V_a}{V_t}\right)$$

For high level injection, n_o and p_o can be neglected

$$(\delta n)(\delta p) \cong n_i^2 \exp\left(\frac{V_a}{V_t}\right)$$

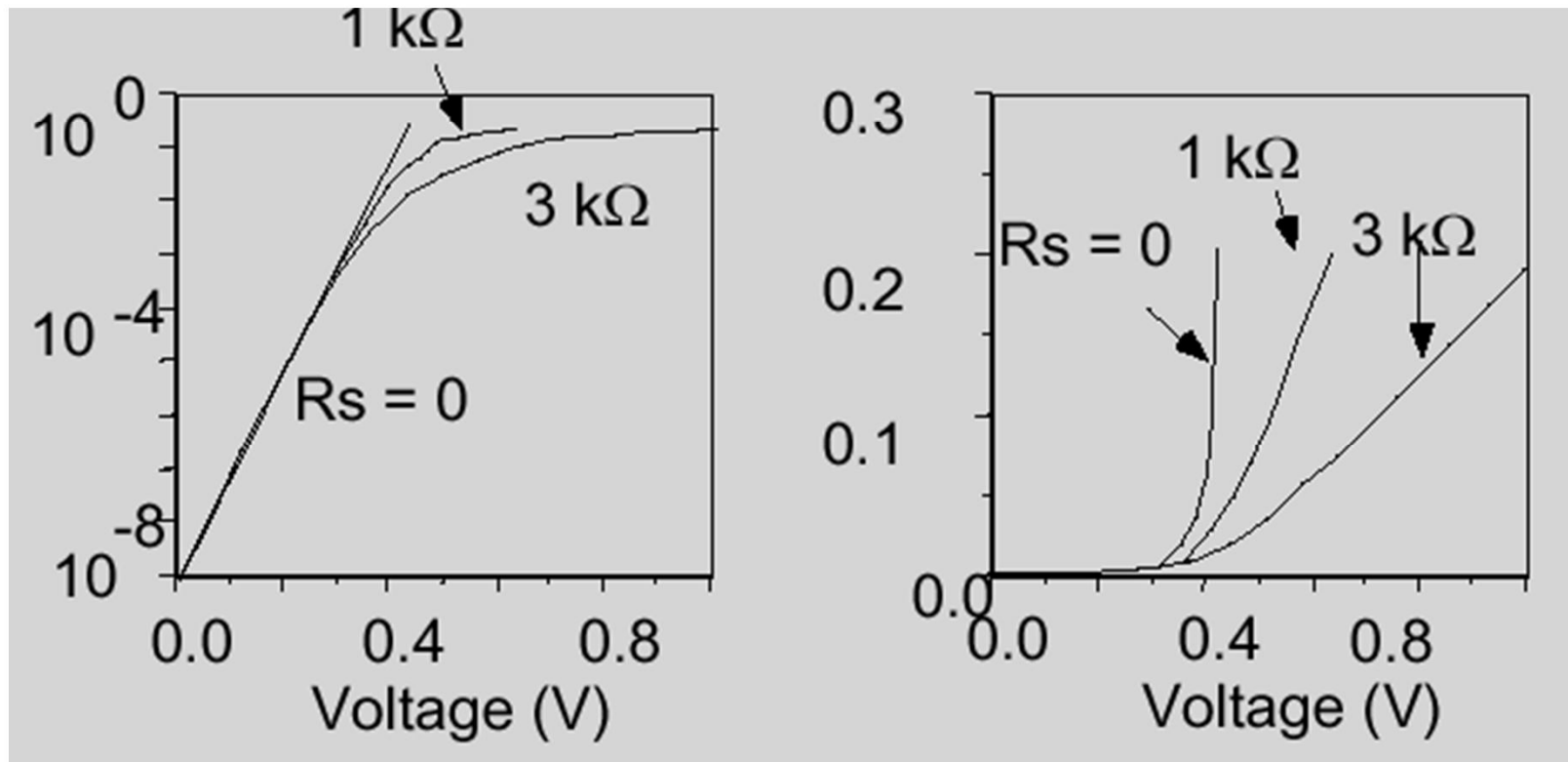
$$\delta n = \delta p \cong n_i \exp\left(\frac{V_a}{2V_t}\right)$$

The diode current is proportional to the excess carrier concentration so that, under high-level injection, we have:

$$I \propto \exp\left(\frac{V_a}{2V_t}\right)$$

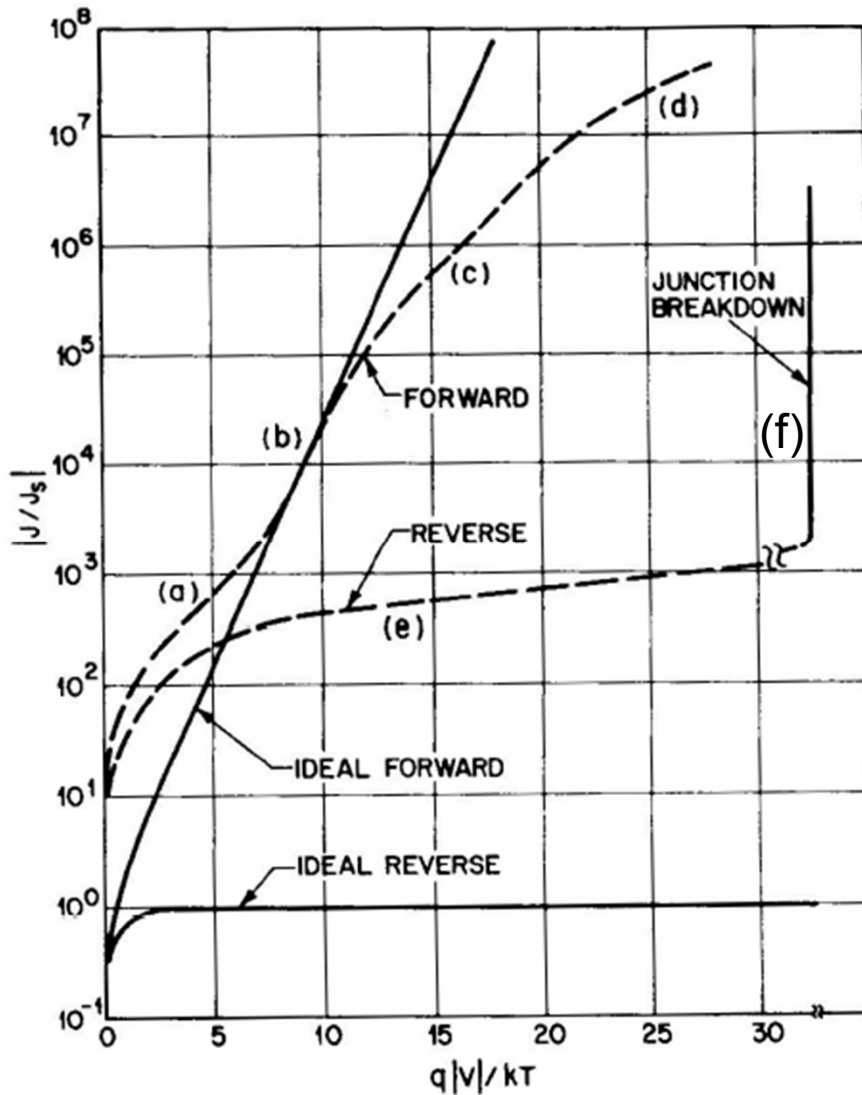
Series Resistance

As the forward bias increases, the devices start to **behave like a resistor**, where the current-voltage relation is given by a simple linear expression. The current is now controlled by the resistance of the n- and p-type regions as well as the contact resistance.



$$I = I_s \left[\exp \left(\frac{V - IR_s}{V_{th}} \right) - 1 \right]$$

Non-Ideal Diode Effects



- (a) Recombination current
- (b) Diffusion current
- (c) High field injection
- (d) Series resistance effect
- (e) Generation current
- (f) Breakdown current

8.3 | SMALL-SIGNAL MODEL OF THE pn JUNCTION

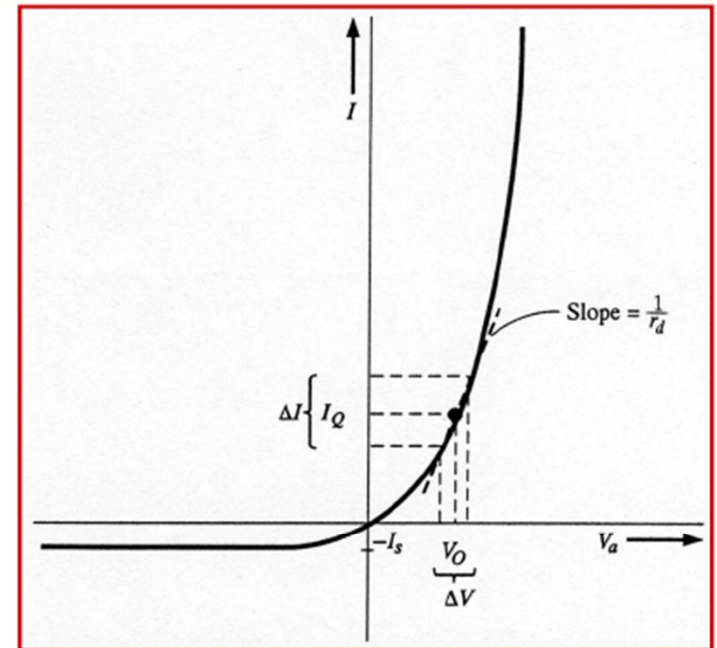
Semiconductor devices with pn junctions are often used in linear amplifier circuits, where sinusoidal signals are superimposed on the dc signals. In such cases, the small-signal characteristics of pn junctions become important.

Ideal current-voltage relationship:

$$J = J_s \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right]$$

Multiplied by the cross-sectional area:

$$I = I_s \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right]$$



A dc voltage V_0 produces a dc current I_Q . A small, low-frequency sinusoidal voltage will produce a small sinusoidal current. The ratio of the sinusoidal current to voltage is called the **incremental conductance**, which is just the slope at the limit of a very small sinusoidal current and voltage.

Diffusion Resistance

$$g_d = \left. \frac{dI}{dV_a} \right|_{V_a=V_0}$$

The reciprocal of the incremental conductance is the **incremental resistance**, also called the **diffusion resistance**:

$$r_d = \left. \frac{dV_a}{dI} \right|_{I=I_Q}$$

If the diode is sufficiently far in the forward-bias region, then the (-1) term can be neglected.

$$g_d = \left. \frac{dI}{dV_a} \right|_{V_a=V_0} = \left(\frac{e}{kT} \right) I_s \exp\left(\frac{eV_0}{kT} \right) \cong \left(\frac{e}{kT} \right) I_Q$$

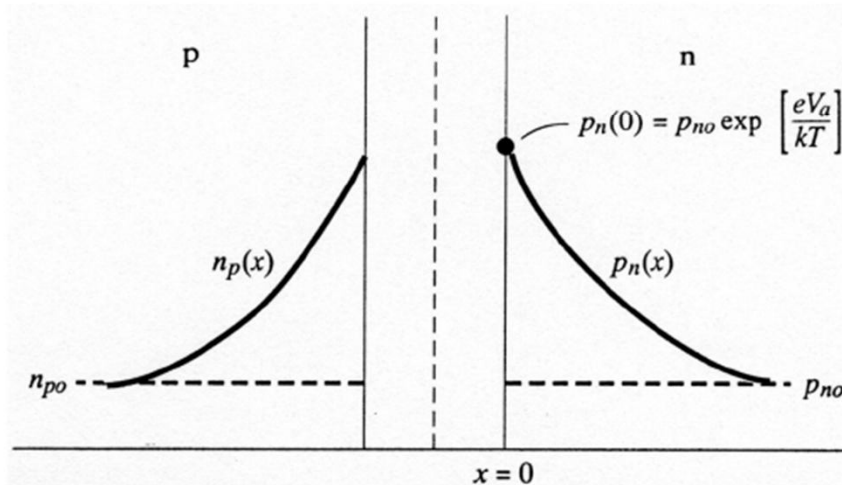
$$r_d = \left(\frac{kT}{e} \right) \frac{1}{I_Q}$$

Admittance

We have considered the junction capacitance under reverse bias. When a pn junction is under forward bias with a small sinusoidal voltage, the small-signal **admittance**, or **impedance**, becomes important. We will derive the admittance from the minority carrier diffusion current.

We consider the n-region. We have derived the minority carrier hole concentration with the origin set at the metallurgical junction:

$$\delta p_n(x) = p_n(x) - p_{n0} = p_{n0} \left[\exp\left(\frac{eV_a}{kT}\right) - 1 \right] \exp\left(\frac{x_n - x}{L_p}\right) \quad (x > x_n)$$



We set the origin at the edge of the depletion region for convenience.

$$p_n(0) = p_{n0} \exp\left(\frac{eV_a}{kT}\right)$$

Admittance

Let $V_a = V_0 + v_1(t)$

V_0 : the dc quiescent voltage
 $v_1(t)$: the ac signal voltage

We obtain

$$p_n(x=0) = p_{n0} \exp\left[\frac{e(V_0 + v_1(t))}{kT}\right] = p_n(0, t)$$

Let

$$p_{dc} = p_{n0} \exp\left(\frac{eV_0}{kT}\right)$$

Then

$$p_n(0, t) = p_{dc} \exp\left(\frac{ev_1(t)}{kT}\right)$$

If we assume

$$|v_1(t)| \ll \left(\frac{kT}{e}\right) = V_t$$

Then

$$p_n(0, t) \cong p_{dc} \left(1 + \frac{v_1(t)}{V_t}\right)$$

Admittance

If $v_1(t)$ is a sinusoidal signal, we obtain

$$p_n(0, t) = p_{dc} \left(1 + \frac{\hat{V}_1}{V_t} e^{j\omega t} \right)$$

Amplitude



In the neutral n-region ($x > 0$), the electric field is assumed to be zero. The behavior of the excess minority carrier holes is determined by:

$$D_p \frac{\partial^2(\delta p_n)}{\partial x^2} - \frac{\delta p_n}{\tau_{p0}} = \frac{\partial(\delta p_n)}{\partial t}$$

We expect the solution for δp_n to be of the form of a sinusoidal function superimposed on the dc solution:

$$\delta p_n(x, t) = \delta p_0(x) + p_1(x) e^{j\omega t}$$

$\delta p_0(x)$ is the dc excess carrier concentration and $p_1(x)$ is the magnitude of the ac component.

Admittance

Substitute $\delta p_n(x,t)$ into the differential equation, we obtain

$$D_p \left\{ \frac{\partial^2 (\delta p_0(x))}{\partial x^2} + \frac{\partial^2 p_1(x)}{\partial x^2} e^{j\omega t} \right\} - \frac{\delta p_0(x) + p_1(x) e^{j\omega t}}{\tau_{p0}} = j\omega p_1(x) e^{j\omega t}$$

Combine time-independent and time-dependent terms:

$$\left\{ D_p \frac{\partial^2 (\delta p_0(x))}{\partial x^2} - \frac{\delta p_0(x)}{\tau_{p0}} \right\} + \left\{ D_p \frac{\partial^2 p_1(x)}{\partial x^2} - \frac{p_1(x)}{\tau_{p0}} - j\omega p_1(x) \right\} e^{j\omega t} = 0$$

||
0

$$D_p \frac{d^2 p_1(x)}{dx^2} - \frac{p_1(x)}{\tau_{p0}} - j\omega p_1(x) = 0$$

Because

$$L_p^2 = D_p \tau_{p0}$$

therefore

$$\frac{d^2 p_1(x)}{dx^2} - \frac{(1 + j\omega \tau_{p0})}{L_p^2} p_1(x) = 0$$

Admittance

Let
$$C_p^2 = \frac{1 + j\omega\tau_{p0}}{L_p^2}$$

We obtain

$$\frac{d^2 p_1(x)}{dx^2} - C_p^2 p_1(x) = 0$$

The general solution is

$$p_1(x) = K_1 e^{-C_p x} + K_2 e^{+C_p x}$$

From the boundary condition of $p_1(x \rightarrow +\infty) = 0$, $K_2 = 0$. Then

$$p_1(x) = K_1 e^{-C_p x}$$

Apply the boundary condition at $x = 0$, we obtain

$$p_1(0) = K_1 = p_{dc} \left(\frac{\hat{V}_1}{V_t} \right)$$

Admittance

The hole diffusion current density can be calculated at $x = 0$:

$$J_p = -eD_p \left. \frac{\partial p_n}{\partial x} \right|_{x=0}$$

The derivative of the hole concentration is just the derivative of the excess hole concentration:

$$J_p = -eD_p \left. \frac{\partial(\delta p_n)}{\partial x} \right|_{x=0} = -eD_p \left. \frac{\partial(\delta p_0(x))}{\partial x} \right|_{x=0} - eD_p e^{j\omega t} \left. \frac{\partial p_1(x)}{\partial x} \right|_{x=0}$$

$$J_p = J_{p0} + j_p(t)$$

$$J_{p0} = -eD_p \left. \frac{\partial(\delta p_0(x))}{\partial x} \right|_{x=0} = \frac{eD_p p_{n0}}{L_p} \left[\exp\left(\frac{eV_0}{kT}\right) - 1 \right]$$

The dc component of the hole diffusion current density is exactly the same as in the ideal I - V relation.

Admittance

The sinusoidal component of the diffusion current density is found from:

$$j_p(t) = -eD_p e^{j\omega t} \left. \frac{\partial p_1(x)}{\partial x} \right|_{x=0} = eD_p C_p p_{dc} \left(\frac{\hat{V}_1}{V_t} \right) e^{j\omega t} e^{-c_p x} \Big|_{x=0}$$

The current density phasor

$$j_p(t) = eD_p C_p p_{dc} \left(\frac{\hat{V}_1}{V_t} \right) e^{j\omega t} = \hat{J}_p e^{j\omega t}$$


The ac hole current phasor is

$$\hat{I}_p = A \hat{J}_p = eAD_p C_p p_{dc} \left(\frac{\hat{V}_1}{V_t} \right) = \frac{eAD_p p_{dc}}{L_p} \sqrt{1 + j\omega\tau_{p0}} \left(\frac{\hat{V}_1}{V_t} \right)$$

If we define

$$I_{p0} = \frac{eAD_p p_{dc}}{L_p} = \frac{eAD_p p_{n0}}{L_p} \exp\left(\frac{eV_0}{kT}\right)$$

Admittance

Then

$$\hat{I}_p = I_{p0} \sqrt{1 + j\omega\tau_{p0}} \left(\frac{\hat{V}_1}{V_t} \right)$$

The same analysis for the minority carrier electrons in the p-region gives

$$\hat{I}_n = I_{n0} \sqrt{1 + j\omega\tau_{n0}} \left(\frac{\hat{V}_1}{V_t} \right) \quad \text{where} \quad I_{n0} = \frac{eAD_n n_{p0}}{L_n} \exp\left(\frac{eV_0}{kT}\right)$$

The total ac current phasor is the sum of the minority carrier hole and electron current phasor. The **pn junction admittance** is the total ac current phasor divided by the ac voltage phasor.

$$Y = \frac{\hat{I}}{\hat{V}_1} = \frac{\hat{I}_p + \hat{I}_n}{\hat{V}_1} = \left(\frac{1}{V_t} \right) \left[I_{p0} \sqrt{1 + j\omega\tau_{p0}} + I_{n0} \sqrt{1 + j\omega\tau_{n0}} \right]$$

Admittance

The equation describing the pn junction admittance is complicated. If we assume that the frequency of the ac signal is not too large:

$$\omega\tau_{p0} \ll 1 \quad \text{and} \quad \omega\tau_{n0} \ll 1$$

then

$$\sqrt{1 + j\omega\tau_{p0}} \approx 1 + \frac{j\omega\tau_{p0}}{2} \quad \text{and} \quad \sqrt{1 + j\omega\tau_{n0}} \approx 1 + \frac{j\omega\tau_{n0}}{2}$$

Substituting these equations into the admittance equation yields:

$$Y = \left(\frac{1}{V_t} \right) \left[I_{p0} \left(1 + \frac{j\omega\tau_{p0}}{2} \right) + I_{n0} \left(1 + \frac{j\omega\tau_{n0}}{2} \right) \right]$$

Combine the real and imaginary portions:

$$Y = \left(\frac{1}{V_t} \right) (I_{p0} + I_{n0}) + j\omega \left\{ \left(\frac{1}{2V_t} \right) [I_{p0}\tau_{p0} + I_{n0}\tau_{n0}] \right\}$$

$$Y = g_d + j\omega C_d$$

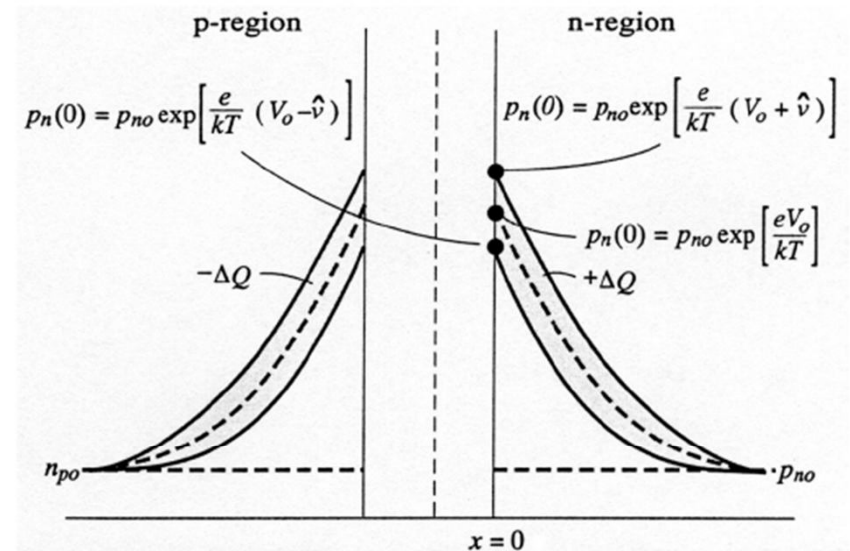
Admittance

$$g_d = \left(\frac{1}{V_t} \right) (I_{p0} + I_{n0}) = \frac{I_Q}{V_t}$$

I_Q is the dc bias current. g_d is called the **diffusion conductance**. It is exactly the same as we obtained previously.

$$C_d = \left(\frac{1}{2V_t} \right) (I_{p0}\tau_{p0} + I_{n0}\tau_{n0})$$

The parameter C_d is called the **diffusion capacitance**.



The ΔQ is alternately being charged and discharged through the junction as the voltage changes. The change in the stored minority carrier charge as a function of the change in voltage is the diffusion capacitance.

Admittance

Example: consider a pn junction with $N_a \gg N_d$ so that $p_{n0} \gg n_{p0}$. This assumption implies that $I_{p0} \gg I_{n0}$. Let $T = 300$ K, $\tau_{p0} = 10^{-7}$ s, and $I_{p0} = I_Q = 1$ mA. Calculate the diffusion capacitance and diffusion resistance.

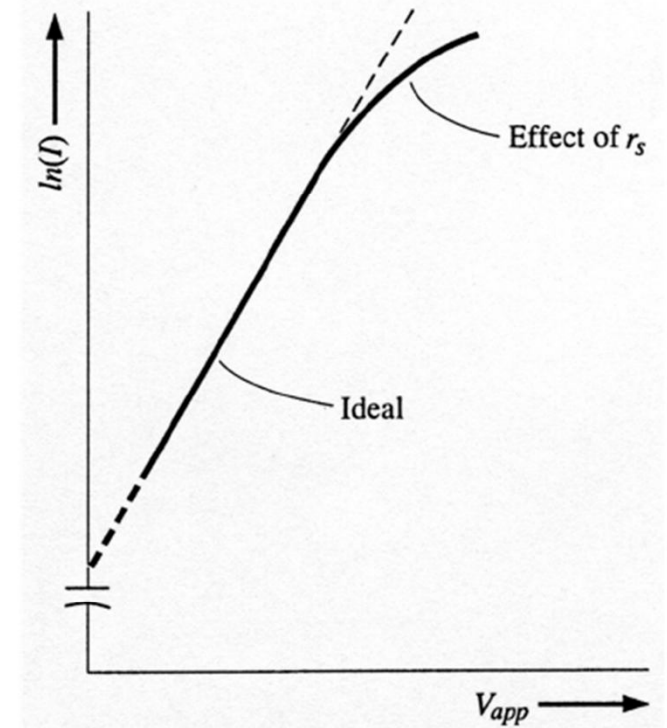
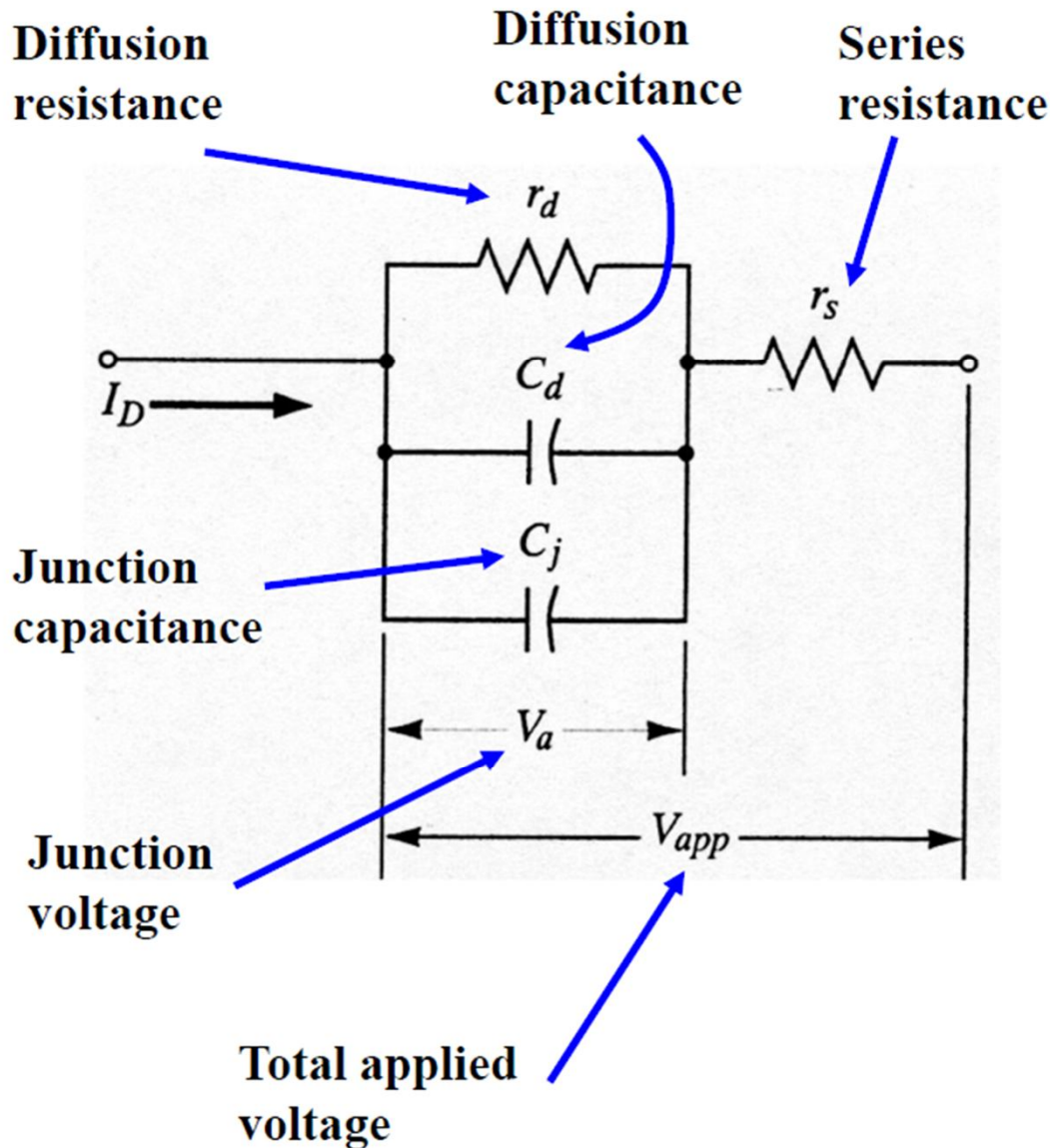
$$C_d = \left(\frac{1}{2V_t} \right) (I_{p0}\tau_{p0} + I_{n0}\tau_{n0}) \cong \left(\frac{1}{2V_t} \right) (I_{p0}\tau_{p0})$$

$$C_d = \frac{1}{(2)(0.0259)} (10^{-3})(10^{-7}) = 1.93 \times 10^{-9} \text{ F}$$

$$r_d = \frac{V_t}{I_Q} = \frac{0.0259}{0.001} = 25.9 \text{ ohms}$$

The diffusion capacitance of a forward-biased pn junction is three to four orders of magnitude larger than the junction capacitance of the reverse-biased pn junction.

Equivalent Circuit



$$V_{app} = V_a + I r_s$$

Reading Materials

D. A. Neamen, “Semiconductor Physics and Devices: Basic Principles”, Fourth edition, Chapter 8, “The pn Junction Diode”.